



Q1 In a database, a rule is defined as ($P1$ and $P2$) or $P3$? $R1(0.8)$ and $R2(0.3)$, where $P1, P2, P3$ are premises and $R1, R2$ are conclusions of rules with certainty factors (CF) 0.8 and 0.3 respectively. If any running program has produced $P1, P2, P3$ with CF as 0.5, 0.8, 0.2 respectively, find the CF of results on the basis of premises.

- (A) $CF(R1 = 0.8), CF(R2 = 0.3)$
 (B) $CF(R1 = 0.40), CF(R2 = 0.15)$
 (C) $CF(R1 = 0.15), CF(R2 = 0.35)$
 (D) $CF(R1 = 0.8), CF(R2 = 0.35)$

Q2 There are three boxes. First box has 2 white, 3 black and 4 red balls. Second box has 3 white, 2 black and 2 red balls. Third box has 4 white, 1 black and 3 red balls. A box is chosen at random and 2 balls are drawn out of which 1 is white, and 1 is red. What is the probability that the balls came from first box?

- (A) 0.237 (B) 0.723
 (C) 0.18 (D) 0.452

Q3 Consider a memory system having address spaced at a distance of m , T = Bank cycle time and n number of banks, then the average data access time per word access in synchronous organization is

- (A) $t = \begin{cases} m \cdot \frac{T}{n} & \text{for } m \ll n \\ T & \text{for } m \gg n \end{cases}$
 (B) $t = \begin{cases} T/n & \text{for } m \ll n \\ T & \text{for } m \gg n \end{cases}$
 (C) $t = \begin{cases} m \cdot T & \text{for } m \ll n \\ T & \text{for } m \gg n \end{cases}$
 (D) $t = \begin{cases} m \cdot T & \text{for } m \ll n \\ m/T & \text{for } m \gg n \end{cases}$

Q4 For multiprocessor system, interconnection network - cross bar switch is an example of

- (A) Non blocking network
 (B) Blocking network
 (C) That varies from connection to connection
 (D) Recurrent network

Q5 The representation of 4 bit code 1101 into 7 bit, even parity Hamming code is

- (A) (1010101) (B) (1111001)
 (C) (1011101) (D) (1110000)

Q6 The number of gate inputs, required to realize expression $ABC + A\bar{B}CD + E\bar{F} + AD$ is

- (A) 12 (B) 13
 (C) 14 (D) 15

Q7 Consider a logic gate circuit, with 8 input lines ($D_0, D_1, \dots, D_7$) and 3 output lines (A_0, A_1, A_2) specified by following operations

$$A_2 = D_4 + D_5 + D_6 + D_7$$

$$A_1 = D_2 + D_3 + D_6 + D_7$$

$$A_0 = D_1 + D_3 + D_5 + D_0$$

Where + indicates logical OR operation. This circuit is

- (A) 3×8 multiplexer
 (B) Decimal to BCD converter
 (C) Octal to Binary encoder
 (D) Priority encoder

Q8 The total storage capacity of a floppy disk having 80 tracks and storing 128 bytes/sector is 163,840 bytes. How many sectors does this disk have?

- (A) 2^7 (B) 2048
 (C) 4K (D) 16

Q9 In a cache memory, if address has 9 bits in Tag field and 12 bits in index field, the size of main memory and cache memory would be respectively

- (A) 2K, 4K (B) 1024K, 2K
 (C) 4K, 2048K (D) 2048K, 4K

Q10 Consider the primal problem :

$$\text{Maximize } z = 5x_1 + 12x_2 + 4x_3$$

$$\text{Subject to } x_1 + 2x_2 + x_3 = 10$$

$$2x_1 - x_2 + 3x_3 = 8$$

$$x_1, x_2, x_3 \geq 0$$

its dual problem is

$$\text{Minimize } w = 10y_1 + 8y_2$$

$$\text{Subject to } y_1 + 2y_2 \geq 5$$

$$2y_1 - y_2 \geq 12$$

$$y_1 + 3y_2 \geq 4$$

Which of the following is correct?

- (A) $y_1 \geq 0, y_2$ unrestricted
 (B) $y_1 \geq 0, y_2 \geq 0$
 (C) y_1 is unrestricted, $y_2 \geq 0$
 (D) y_1 is unrestricted, y_2 restricted

Q11 The logic expression

$$(\bar{P} \wedge Q) \vee (P \wedge \bar{Q}) \vee (P \wedge Q)$$

is equivalent to

- (A) $\bar{P} \vee Q$
 (B) $P \vee \bar{Q}$
 (C) $P \vee Q$
 (D) $\bar{P} \vee \bar{Q}$

Q12 The reduced grammar equivalent to the grammar, whose production rules are given below, is

$$S \rightarrow AB \mid CA$$

$$B \rightarrow BC \mid AB$$

$$A \rightarrow a$$

$$C \rightarrow aB \mid b$$

$$(A) S \rightarrow CA, A \rightarrow a, C \rightarrow b$$

$$(B) S \rightarrow CA \mid B, B \rightarrow BC \mid B, A \rightarrow a, C \rightarrow aB \mid b$$



- (C) $S \rightarrow CA|B, B \rightarrow BC, A \rightarrow a, C \rightarrow aB|b$
 (D) $S \rightarrow AB|AC, B \rightarrow BC|BA, A \rightarrow a, C \rightarrow aB|b$

Q13 Consider the production rules of grammar G :

$S \rightarrow AbB$
 $A \rightarrow aAb | \lambda$
 $B \rightarrow bB | \lambda$

Which of the following language L is generated by grammar G ?

- (A) $L = \{a^n b^m : n \geq 0, m > n\}$
 (B) $L = \{a^n b^m : n \geq 0, m \geq 0\}$
 (C) $L = \{a^n b^m : n \geq m\}$
 (D) $L = \{a^n b^m : n \geq m, m > 0\}$

Q14 Consider the language

$L = \{a^n b^m : n \geq 4, m \leq 3\}$

Which of the following regular expression represents language L ?

- (A) $aaaa^*(\lambda + b + bb + bbb)$
 (B) $aaaaa^*(b + bb + bbb)$
 (C) $aaaaa^*(\lambda + b + bb + bbb)$
 (D) $aaaaa^*(b + bb + bbb)$

Q15 Consider $L = \{ab, aa, baa\}$

Which of the following string is NOT in L^* ?

- (A) $baaaaaabaaaa$ (B) $abaabaaabaa$
 (C) $aaaabaaaa$ (D) $baaaaaabaa$

Q16 Consider the following NPDA

$= (\{q_0, q_1, q_f\}, \{a, b\}, \{1, z\}, \delta, q_0, z, \{q_f\})$

$\delta(q_0, \lambda, z) = \{(q_f, z)\}$

$\delta(q_0, a, z) = \{(q_1, 11z)\}$

$\delta(q_1, a, 1) = \{(q_1, 111)\}$

$\delta(q_1, b, 1) = \{(q_1, \lambda)\}$

$\delta(q_1, \lambda, z) = \{(q_f, z)\}$

Which of the following Language L is accepted by NPDA?

- (A) $L = \{a^{2n} b^n : n \geq 0\}$ (B) $L = \{a^n b^{2n} : n \geq 0\}$
 (C) $L = \{a^{2n} b^n : n > 0\}$ (D) $L = \{a^n b^{2n} : n > 0\}$

Q17 Hidden surface removal problem with minimal 3D pipeline can be solved with

- (A) Painter's algorithm
 (B) Window Clipping algorithm
 (C) Brute force rasterization algorithm
 (D) Flood fill algorithm

Q18 Using 'RSA' algorithm, if $p = 13, q = 5$ and $e = 7$, the value of d and cipher value of '6' with (e, n) key are-

- (A) 7,4 (B) 7,1
 (C) 7,46 (D) 55,1

Q19 The condition $num! = 65$ cannot be replaced by-

- (A) $num > 65 || num < 65$
 (B) $!(num == 65)$
 (C) $num - 65$
 (D) $!(num - 65)$

Q20 Pointers cannot be used to-

- (A) find the address of a variable in memory
 (B) reference value directly
 (C) simulate call by reference
 (D) manipulate dynamic data structure

Q21 Which mechanism in XML allows organizations to specify globally unique names as element tags in documents?

- (A) root (B) header
 (C) schema (D) namespace

Q22 If an operating system does not allow a child process to exist when the parent process has been terminated, this phenomenon is called as -

- (A) Threading
 (B) Cascading termination
 (C) Zombie termination
 (D) Process killing

Q23 What is called Journalling in Linux operating system?

- (A) Process scheduling
 (B) File saving as transaction
 (C) A type of thread
 (D) An editor

Q24 This transformation is called

$$\begin{bmatrix} \bar{x} \\ \bar{y} \\ \bar{z} \\ \bar{w} \end{bmatrix} = \begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ e & f & g & h \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

- (A) Scaling (B) Shear
 (C) Homography (D) Steganography

Q25 RAD software process model stands for-

- (A) Rapid Application Development
 (B) Relative Application Development
 (C) Rapid Application Design
 (D) Recent Application Development

Q26 If every requirement can be checked by a cost-effective process, then SRS is called

- (A) Verifiable (B) Tractable
 (C) Modifiable (D) Complete

Q27 Fault base testing technique is

- (A) Unit testing
 (B) Beta testing
 (C) Stress testing
 (D) Mutation testing



- Q28** Alpha and Beta testing are forms of
 (A) White-Box Testing
 (B) Black - Box Testing
 (C) Acceptance Testing
 (D) System Testing

- Q29** The process to gather the software requirements from client, analyze and document is known as -
 (A) Software Engineering Process
 (B) User Engineering Process
 (C) Requirement Elicitation Process
 (D) Requirement Engineering Process

- Q30** Size and complexity are a part of
 (A) People Metrics
 (B) Project Metrics
 (C) Process Metrics
 (D) Product Metrics

Directions (31-35) Read the following passage and answer the given questions.

Read the following and Answer the questions:

Consider the relational schema of sailors S.

Reserves R and Boats B.

Table 1: Sailors S

Sid	Sname	Rating	Age
22	Dustin	7	45.0
29	Brutus	1	33.0
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35.5
64	Horatio	7	35.5
71	Zorba	10	16.0
74	Horatio	9	35.0
85	Art	3	25.5
95	Bob	3	63.5

Table 2 : Reserves R

Sid	Bid	day
22	101	10/10/98
22	102	10/10/98
22	103	10/8/98
22	104	10/7/98
31	102	11/10/98
31	103	11/6/98
31	104	11/12/98
64	101	9/5/98
64	102	9/8/98
74	103	9/8/98

Table 3 : Boats B

Bid	Bname	Color
101	Interlake	blue
102	Interlake	red
103	Clipper	green
104	Marine	red

- Q31** Which of the following relational algebra query computes the Sid's of sailors with age over 20 who have not reserved a red boat?

- (A) $\pi_{sid} (\delta_{age > 20} Sailors) - \pi_{sid} ((\delta_{color = red} Boats) \bowtie Reserves \bowtie \Delta Sailors)$
 (B) $\pi_{sid} ((\delta_{color = 'red'} \wedge age = 20 (Boats \bowtie Sailors \bowtie Reserves)))$
 (C) $\pi_{sid} ((\delta_{age < 20} Sailors) - \pi_{sid} ((\delta_{color = 'red'} Boats) \bowtie Reserves \bowtie Sailors))$
 (D) $\pi_{sid} ((\delta_{age > 20} Sailors) \wedge \pi_{sid} ((\delta_{color = 'red'} Boats) \bowtie Reserves \bowtie Sailors))$

- Q32** Which of the following relational algebra query/queries computes/compute the names of sailors who have reserved a red boat?

- Q1 $\pi_{sname} ((\delta_{color = 'red'} Boats) \bowtie Reserves \bowtie Sailors)$
 Q2 $\pi_{sname} (\pi_{sid} ((\pi_{bid} \delta_{color = 'red'} Boats) \bowtie Reserves) \bowtie Sailors)$
 Q3 $\pi_{sname} (\delta_{color = 'red'} Reserves) \bowtie Boats \bowtie Sailors)$
 (A) Both Q1 and Q2 (B) Both Q2 and Q3
 (C) Only Q1 (D) Only Q2

- Q33** Which of the following relational algebra query/queries computes/compute the name of sailors who have reserved boat 103?

- Q1 $\pi_{sname} ((\delta_{bid = 103} Boats) \bowtie Sailors)$
 Q2 $\pi_{sname} ((\delta_{bid = 103} (Reserves \bowtie Sailors)))$
 Q3 $\pi_{sname} ((\delta_{bid = 103} Reserves) \bowtie Sailors)$
 (A) Both Q1 and Q3 (B) Both Q2 and Q3
 (C) Only Q3 (D) Only Q2

- Q34** Which of the following relational algebra query computes the names of sailor who have reserved all boats?

- (A) $\rho(Tempsids, (\pi_{bid} Reserves) / \pi_{bid} Boats) \pi_{sname} ((Tempids) \bowtie Sailors)$
 (B)



- $\rho(\text{Tempsids} \bowtie (\pi_{\text{sid}, \text{bid}} \text{Reserves})$
 $/ \pi_{\text{bid}} \text{Boats}) \pi_{\text{sname}}((\text{Tempsids})$
 $\bowtie \text{Sailors})$
 (C) $\rho(\text{Tempsids} \bowtie (\pi_{\text{s id}} \text{Sailors})$
 $/ \pi_{\text{bid}} \text{Boats}) \pi_{\text{sname}}((\text{Tempsids})$
 $\bowtie \text{Sailors})$
 (D) $\rho(\text{Tempsids} \bowtie (\pi_{\text{sid}} \text{Reserves})$
 $/ \pi_{\text{bid}} \text{Boats}) \pi_{\text{sname}}((\text{Tempsids})$
 $\bowtie \text{Boats})$

Q35 Which of the following relational algebra query computes the names of sailors who have reserved a red and a green boat?

- (A) $\rho(\text{Tempred},$
 $\pi_{\text{sid}}((\delta_{\text{color}} = \text{'red'}} \text{Boats})$
 $\bowtie \text{Reserves})) \cdot \rho(\text{Tempgreen},$
 $\pi_{\text{sid}}((\delta_{\text{color}} = \text{'green'}} \text{Boats})$
 $\bowtie \text{Reserves})) \cdot \pi_{\text{sname}}(\text{Tempred}$
 $\cap \text{Tempgreen}) \bowtie \text{Sailors}$
 (B) $\rho(\text{Tempboats}$
 $2(\delta_{\text{color}} = \text{'red'}} \text{Boats})$
 $\cap (\delta_{\text{color}} = \text{'green'}} \text{Boats}) \pi_{\text{sname}}$
 $(\text{Tempboats} 2 \bowtie \text{Reserves})$
 (C) $\pi_{\text{sname}}((\delta_{\text{color}} = \text{'red'}} \text{Boats})$
 $\cap (\delta_{\text{color}}$
 $= \text{'green'}} \text{Boats})) (\text{Tempboats} 2$
 $\bowtie \text{Sailors} \bowtie \text{Reserves})$
 (D) $\rho(\text{Tempboats} 2$
 $\cdot (\delta_{\text{color}} = \text{'red'}} \wedge \text{'color'} = \text{'green'}} \text{Boats})$
 $(\pi_{\text{sname}}(\text{Tempboats} 2 \bowtie \text{Sailors}))$

Directions (36-40) Read the following passage and answer the given questions.

Based on the following passage, answer the Questions:

A 3000 km long trunk operates at 1.536 mbps and is used to transmit 64 bytes frames and uses sliding window protocol. The propagation speed is 6μsec/km.

- Q36** The transmission and propagation delays are respectively
 (A) $T_t = 333.33\mu\text{sec}$, $T_p = 18000\mu\text{sec}$
 (B) $T_t = 300\mu\text{sec}$, $T_p = 15360\mu\text{sec}$
 (C) $T_t = 33.33\mu\text{sec}$, $T_p = 1800\mu\text{sec}$
 (D) $T_t = 1800\mu\text{sec}$, $T_p = 33.33\mu\text{sec}$
- Q37** The minimum number of bits required in the sequence number field of the packet is
 (A) 6 bits (B) 7 bits
 (C) 5 bits (D) 4 bits
- Q38** The sender window size to get the maximum efficiency is
 (A) 108 (B) 109
 (C) 55 (D) 56

- Q39** If only 6 bits are reserved for sequence number field, then the efficiency of the system is
 (A) 0.587 (B) 0.875
 (C) 0.578 (D) 0.50

- Q40** The maximum achievable throughput is
 (A) 0.768 (B) 0.678
 (C) 0.901 (D) 0.887

- Q41** Which Metrics are derived by normalizing quality and/or productivity measures by considering the size of the software that has been produced?
 (A) Function - Oriented Metrics
 (B) Function - Point Metrics
 (C) Line of Code Metrics
 (D) Size Oriented Metrics

- Q42** The model in which the requirements are implemented by its category is
 (A) Evolutionary Development Model
 (B) Waterfall Model
 (C) Prototyping Model
 (D) Iterative Enhancement Model

- Q43** Which of the following is an indirect measure of product?
 (A) Quality (B) Complexity
 (C) Reliability (D) All of these

- Q44** Modules X and Y operate on the same input and output, then the cohesion is
 (A) Logical cohesion
 (B) Sequential cohesion
 (C) Procedural cohesion
 (D) Communicational cohesion

- Q45** Which mode is a block cipher implementation as a self synchronizing stream cipher?
 (A) Cipher Block Chaining Mode
 (B) Cipher Feedback Mode
 (C) Electronic Codebook Mode
 (D) Output Feedback Mode

- Q46** Which one is a connectionless transport-layer protocol that belongs to the Internet protocol family?
 (A) Transmission Control Protocol (TCP)
 (B) User Datagram Protocol (UDP)
 (C) Routing Protocol (RP)
 (D) Datagram Control Protocol (DCP)

- Q47** Consider an error-free 64 kbps satellite channel used to send 512-byte data frames in one direction with very short acknowledgments coming back the other way. What is the maximum throughput for a window size of 15?
 (A) 32 kbps (B) 48 kbps
 (C) 64 kbps (D) 70 kbps



Q48 A classless address is given as 167.199.170.82/27.
The number of addresses in the network is
(A) 64 addresses (B) 32 addresses
(C) 28 addresses (D) 30 addresses

Q49 Which layer divides each message into packets at the source and re-assembles them at the destination?
(A) Network layer
(B) Transport layer
(C) Data link layer
(D) Physical layer

Q50 A 4-stage pipeline has a stage delay of 150,120,160 and 140 ns respectively. Registers that are used between the stages have a delay of 5 ns. Assuming constant locking rate, the total time required to process 1000 data items on this pipeline is
(A) 160.5 ms (B) 165.5 ms
(C) 120.5 ms (D) 590.5 ms

Q51 Which of the following is correct for the destination address 4A:30:10:21:10:1A?
(A) unicast address
(B) multicast address
(C) broadcast address
(D) unicast and broadcast address

Q52 Assume that $f(n)$ and $g(n)$ are asymptotically positive. Which of the following is correct?
(A) $f(n) = O(g(n))$ and $g(n) = O(h(n)) \Rightarrow f(n) = \omega(h(n))$
(B) $f(n) = \Omega(g(n))$ and $g(n) = \Omega(h(n)) \Rightarrow f(n) = O(h(n))$
(C) $f(n) = o(g(n))$ and $g(n) = o(h(n)) \Rightarrow f(n) = o(h(n))$
(D) $f(n) = \omega(g(n))$ and $g(n) = \omega(h(n)) \Rightarrow f(n) = \Omega(h(n))$

Q53 The solution of the recurrence relation $T(n) = 3T(\frac{n}{4}) + n \log n$ is
(A) $\theta(n^2 \cdot \log n)$
(B) $\theta(n \cdot \log n)$
(C) $\theta(n \cdot \log n)^2$
(D) $\theta(\log \log n)$

Q54 The number of nodes of height h in any n -element heap is atmost:
(A) $\frac{n}{2^{h+1}}$ (B) $\frac{n}{2^{h-1}}$
(C) $\frac{n}{2^h}$ (D) $\frac{n-1}{2^{h-1}}$

Q55 Consider a B-tree of height h minimum degree $t \geq 2$ that contains any n -key, where $n \geq 1$. Which of the following is correct?
(A) $h \geq \log_t \frac{n+1}{2}$ (B) $h \leq \log_t \frac{n+1}{2}$
(C) $h \geq \log_t \frac{n-1}{2}$ (D) $h \leq \log_t \frac{n-1}{2}$

Q56 Which of the following algorithm design approach is used in Quick sort algorithm?
(A) Dynamic programming
(B) Back Tracking
(C) Divide and conquer
(D) Greedy approach

Q57 Consider the hash table of size 11 that uses open addressing with linear probing. Let $h(k) = k \bmod 11$ be the hash function. A sequence of records with keys 43,36,92,87,11,47,11,13,14 is inserted into an initially empty hash table, the bins of which are indexed from 0 to 10. What is the index of the bin into which the last record is inserted?
(A) 8 (B) 7
(C) 10 (D) 4

Q58 Consider the traversal of a tree
Preorder $\rightarrow$ ABCEIFJDGHKL
Inorder $\rightarrow$ EICFJBGDKHLA
Which of the following is correct post order traversal?
(A) EIFJCKGLHDBA (B) FCGKLHDBUAE
(C) FCGKLHDBAEIJ (D) IEJFCGKLHDBA

Q59 How many rotations are required during the construction of an AVL tree if the following elements are to be added in the given sequence? 35, 50, 40, 25, 30, 60, 78, 20, 28
(A) 2 left rotations, 2 right rotations
(B) 2 left rotations, 3 right rotations
(C) 3 left rotations, 2 right rotations
(D) 3 left rotations, 1 right rotation

Q60 Consider the following two lists :

List I	List II
(A) Stack overflow	(I) Software Interrupt
(B) Timer	(II) Internal interrupt
(C) Invalid opcode	(III) External interrupt
(D) Superior call	(IV) Machine check interrupt

(A) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
(B) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
(C) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
(D) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

Q61 Let $R(ABCDEFGH)$ be a relation schema and F be the set of dependencies
 $F = \{A \rightarrow B, ABCD \rightarrow E, EF \rightarrow G, EF \rightarrow H, ACDF \rightarrow EG\}$. The minimal cover of a set of functional dependencies is
(A) $A \rightarrow B, ACD \rightarrow E, EF \rightarrow G$, and $EF \rightarrow H$
(B) $A \rightarrow B, ACD \rightarrow E, EF \rightarrow G, EF \rightarrow H$ and $ACDF \rightarrow G$



- (C) $A \rightarrow B, ACD \rightarrow E, EF \rightarrow G, EF \rightarrow H$
and $ACDF \rightarrow E$
- (D) $A \rightarrow B, ABCD \rightarrow E, EF \rightarrow H$, and
 $EF \rightarrow G$

Q62 A trigger is

- (A) A statement that enables to start DBMS.
- (B) A statement that is executed by the user when debugging an application program.
- (C) A condition the system tests for the validity of the database user.
- (D) A statement that is executed automatically by the system as a side effect of modification to the database.

Q63 For the following page reference string 4,3,2,1,4,3,5,4,3,2,1,5, the number of page faults that occur in the least Recently Used (LRU) page replacement algorithm with frame size 3 is:

- (A) 6 (B) 8
- (C) 10 (D) 12

Q64 A magnetic tape drive has a transport speed of 200 inches per second and a recording density of 1600 bytes per inch. The time required to write 600000 of data grouped in 100 characters records with a blocking factor of 10 is

- (A) 1.875 sec
- (B) 2.6251 sec
- (C) 2.0062 sec
- (D) 2.6150 sec

Q65 Consider two lists A and B of three strings on $\{0, 1\}$

X:

List A	List B
1	111
10111	10
10	0

Y:

List A	List B
10	101
011	11
101	011

Which of the following is true?

- (A) Only PCP in X has solution.
- (B) Only PCP in Y has solution.
- (C) PCP in both X and Y has solution.
- (D) PCP neither in X nor in Y has solution.

Q66 Consider the properties of recursively enumerable sets:

- (A) Finiteness

(B) Context Freedom

(C) Emptiness

Which of the following is true?

- (A) Only (A) and (B) are not decidable
- (B) Only (B) and (C) are not decidable
- (C) Only (C) and (A) are not decidable
- (D) All (A), (B) and (C) are not decidable

Q67 Consider the following:

	List I		List II
(A)	Activation record	(I)	Linker Loader
(B)	Location counter	(II)	Garbage Collection
(C)	Reference count	(III)	Subroutine Call
(D)	Address relocation	(IV)	Assembler

Which of the following is correct matching?

- (A) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
- (B) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)
- (C) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
- (D) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

Q68 Consider the following related to Fourth Generation Technique (4GT):

- (A) It controls efforts.
- (B) It controls resources.
- (C) It controls cost of development.
- Choose the correct answer from the options given below:

- (A) (A) and (B) only
- (B) (B) and (C) only
- (C) (C) and (A) only
- (D) All (A), (B) and (C)

Q69 Consider the grammar $S \rightarrow SbS|a$.

Consider the following statements:

The string abababa has

- (A) two parse trees
- (B) two left most derivations
- (C) two right most derivations

Which of the following is correct?

- (A) All (A), (B) and (C) are true
- (B) Only (B) is true
- (C) Only (C) is true
- (D) Only (A) is true

Q70 In a game playing search tree, upto which depth $\alpha - \beta$ pruning can be applied?

- (A) Root (0) level
- (B) 6 level
- (C) 8 level
- (D) Depends on utility value in a breadth first order

Choose the correct answer from the options given below:

- (A) (B) and (C) only



- (B) (A) and (B) only
 (C) (A), (B) and (C) only
 (D) (A) and (D) only

Q71 Consider α, β, γ as logical variables. Identify which of the following represents correct logical equivalence :

- (A) $(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma))$
 (B) $(\alpha \vee \beta) \equiv \neg \alpha \vee \beta$
 (C) $(\alpha \Rightarrow \beta) \equiv (\neg \beta \Rightarrow \neg \alpha)$
 (D) $(\neg(\alpha \vee \beta)) \equiv (\neg \alpha \Rightarrow \neg \beta)$

Choose the correct answer from the options given below :

- (A) (A) and (D) only
 (B) (B) and (C) only
 (C) (A) and (C) only
 (D) (B) and (D) only

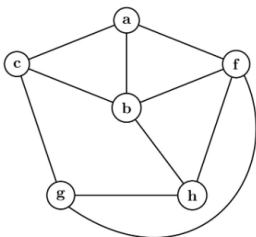
Q72 Let $(\{a, b\}, *)$ be a semigroup, where $a * a = b$.

- (A) $a * b = b * a$
 (B) $b * b = b$

Choose the most appropriate answer from the options given below :

- (A) (A) only true
 (B) (B) only true
 (C) Both (A) and (B) true
 (D) Neither (A) nor (B) true

Q73 Consider the following graph :



For the graph; the following sequences of depth first search (DFS) are given

- (A) abcghf
 (B) abfchg
 (C) abfhgc
 (D) afghbc

Which of the following is correct?

- (A) (A), (B) and (D) only
 (B) (A), (B), (C) and (D)
 (C) (B), (C) and (D) only
 (D) (A), (C) and (D) only

Q74 Let $\epsilon = 0.0005$. and Let R_ϵ be the relation $\{(x, y) \in R^2 : |x - y| < \epsilon\}$, R_ϵ could be interpreted as the relation approximately equal.

- R_ϵ is
 (A) Reflexive
 (B) Symmetric
 (C) transitive

Choose the correct answer from the options given below :

- (A) (A) and (B) only true
 (B) (B) and (C) only true
 (C) (A) and (C) only true
 (D) (A), (B) and (C) true

Q75 In reference to Big data, consider the following database :

- (A) Memcached
 (B) Couch DB
 (C) Infinite graph

Choose the most appropriate answer from the options given below :

- (A) (A) and (B) only
 (B) (B) and (C) only
 (C) (C) and (A) only
 (D) (A), (B) and (C)

Q76 Match List I with List II :

List-I	List-II
A. BCNF	I. It removes multivalued dependency.
B. 3NF	II. It is not always dependency preserving.
C. 2NF	III. It removes transitive transparency.
D. 4NF	IV. It removes partial functional dependency..

Choose the correct answer from the options given below:

- (A) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)
 (B) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)
 (C) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
 (D) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

Q77 Match List I with List II :

List-I	List-II
A. Localization	I. Encapsulation
B. Packaging or binding of a collection of items.	II. Abstraction
C. Mechanism that enables a designer to focus on essential details of a program component.	III. Characteristic of software that indicates in a manner which information is concentrated in the program.
D. Information hiding.	IV. Suppressing the operational details of a program component.

Choose the correct answer from the options given below :



- (A) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
 (B) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)
 (C) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)
 (D) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

Q78 Match List I with List II :

	List I		List II
(A)	Type 0	(I)	Finite Automata
(B)	Type 1	(II)	Turing Machine
(C)	Type 2	(III)	Linear bound Automata
(D)	Type 3	(IV)	Pushdown Automata

Choose the correct answer from the options given below :

- (A) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
 (B) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
 (C) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
 (D) (A)-(II), (B)-(III), (C)-(II), (D)-(IV)

Q79 Match List I with List II:

	List-I		List-II
(A)	Ontological Engineering	(I)	Organizing subclass relations
(B)	Taxonomy Hierarchy	(II)	Organizing knowledge into categories and sub-categories
(C)	Inheritance	(III)	Attaches a number with each possibility
(D)	Probability Modes	(IV)	Representing concepts, events, time, and physical concepts of different domains

Choose the correct answer from the options given below :

- (A) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
 (B) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
 (C) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)
 (D) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

Q80 Match List I with List II :

List I	List II
(A) Least frequently used	(I) Memory is distributed among processors
(B) Critical Section	(II) Page replacement policy in cache memory
(C) Loosely coupled multiprocessor system	(III) Program section that once begun must complete execution before another processor accesses the same shared resource
(D) Distributed operating system	(IV) O/S routines are distributed among

organization	available processors
--------------	----------------------

Choose the correct answer from the options given below :

- (A) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)
 (B) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
 (C) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
 (D) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

Q81 Match List I with List II :

List I	List II
(A) Firmware	(I) Number of logical records into physical blocks
(B) Batch file	(II) ASCII format
(C) Packing	(III) Resource allocation
(D) Banker's Algorithm	(IV) ROM

Choose the correct answer from the options given below :

- (A) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
 (B) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)
 (C) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)
 (D) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

Q82 Match List I with List II:

List I	List II
(A) DES	(I) Key size - 256
(B) AES	(II) Key size - 1024
(C) 3 DES	(III) Key size - 56
(D) RSA	(IV) Key size - 168

Choose the correct answer from the options given below :

- (A) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
 (B) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
 (C) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
 (D) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

Q83 Match List I with List II :

List I	List II
(A) BIND	(I) Block the caller until a connection attempt arrives
(B) LISTEN	(II) Give a local address to a socket
(C) ACCEPT	(III) Show willingness to accept connections
(D) SOCKET	(IV) Create a new point

Choose the correct answer from the options given below :

- (A) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)
 (B) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
 (C) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
 (D) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

Q84 Match List I with List II :

List I	List II
--------	---------



(A) Physical layer	(I) Routing of the signals, divide the outgoing message into packets, act as network controller for routing data
(B) Data link layer	(II) Make and break connections, define voltages and data rates, convert data bits into electrical signal
(C) Network layer	(III) Synchronization, error detection, and correction, assemble outgoing message into frames
(D) Presentation layer	(IV) It works as a translating layer

Choose the correct answer from the options given below :

- (A) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
 (B) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
 (C) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)
 (D) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

Q85 Consider the following algorithms and their running times :

List I	List II
(A) Breadth First Search	(I) $\theta(v+E)$
(B) Rabin-Karp Algorithm	(II) $O(v+E)$
(C) Depth-First Search	(III) $\theta((n-m-1)m)$
(D) Heap sort (worst case)	(IV) $O(n^2)$
(E) Quick sort (worst case)	(V) $O(n \log n)$

Which one of the following is correct?

- (A) (A)-(III), (B)-(II), (C)-(I), (D)-(IV), (E)-(V)
 (B) (A)-(II), (B)-(III), (C)-(I), (D)-(IV), (E)-(V)
 (C) (A)-(II), (B)-(III), (C)-(I), (D)-(V), (E)-(IV)
 (D) (A)-(III), (B)-(I), (C)-(II), (D)-(IV), (E)-(V)

Q86 Consider the following:

List I	List II
(A) Stack algorithm	(I) Deadlock
(B) Elevator algorithm	(II) Disk scheduling
(C) Priority scheduling algorithm	(III) Page replacement
(D) Havender's algorithm	(IV) CPU scheduling

Which of the following is the correct matching?

- (A) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)
 (B) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
 (C) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
 (D) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

Q87 Consider the following statements of approximation algorithm :
 Statement I: Vertex-cover is a polynomial time 2-approximation algorithm.

Statement II: TSP-tour is a polynomial time 3-approximation algorithm for travelling salesman problem with the triangle inequality.

Which of the following is correct?

- (A) Statement I true and Statement II false
 (B) Statement I and Statement II true
 (C) Statement I false and Statement II true
 (D) Statement I and Statement II false

Q88 Consider the following statements:

Statement I: Conservative 2 PL is a deadlock-free protocol.

Statement II: Thomas's write rule enforces conflict serializability.

Statement III : Timestamp ordering protocol ensures serializability based on the order of transaction timestamps.

Which of the following is correct?

- (A) Statement I, Statement II true, and Statement III false
 (B) Statement I, Statement III true, and Statement II false
 (C) Statement I, Statement II false, and Statement III true
 (D) Statement I, Statement II, and Statement III true

Q89 Consider the following statements:

Statement I: Composite attributes cannot be divided into smaller subparts.

Statement II: Complex attribute is formed by nesting composite attributes and multivalued attributes in an arbitrary way.

Statement III: A derived attribute is an attribute whose values are computed from other attribute.

Which of the following is correct?

- (A) Statement I, Statement II and Statement III are true
 (B) Statement I true and Statement II, Statement III false
 (C) Statement I, Statement II true and Statement III false
 (D) Statement I false and Statement II, Statement III true

Q90 A top down approach to programming calls for :

Statement I : Working from the general to the specific.

Statement II : Postpone the minor decisions.

Statement III : A systematic approach.

Statement IV: Intermediate coding of the problem

Which of the following is true?



- (A) Statement I only
- (B) Statement I and Statement II only
- (C) Statement I, Statement II and Statement III only
- (D) Statement I, Statement II and Statement IV only

Q91 Consider the following statements:

Statement I: LALR parser is more powerful than canonical LR Parser.

Statement II: SLR parser is more powerful than LALR

Which of the following is correct?

- (A) Statement I true and Statement II false
- (B) Statement I false and Statement II true
- (C) Both Statement I and Statement II false
- (D) Both Statement I and Statement II true

Q92 Consider the following statements about Context Free Language (CFL):

Statement I: CFL is closed under homomorphism.

Statement II: CFL is closed under complement.

Which of the following is correct?

- (A) Statement I is true and Statement II is false
- (B) Statement II is true and Statement I is false
- (C) Both Statement I and Statement II are true
- (D) Neither Statement I nor Statement II is true

Q93 Consider the following in Boolean Algebra

$$X : a \vee (b \wedge (a \vee c)) = (a \vee b) \wedge (a \vee c)$$

$$Y : a \wedge (b \vee (a \wedge c)) = (a \wedge b) \vee (a \wedge c)$$

$$a \vee (b \wedge c) = (a \vee b) \wedge c \text{ is satisfied if}$$

- (A) X is true
- (B) Y is true
- (C) Both X and Y are true
- (D) It does not depend on X and Y

Q94 A good software requirement specification does NOT have the characteristic

- (A) Completeness
- (B) Consistency
- (C) Clarity
- (D) Reliability

Q95 Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A) : $\bar{p}$

Reason (R): $(r \rightarrow \bar{q}, r \vee s, s \rightarrow \bar{q}, p \rightarrow q)$

In the light of the above statements, choose the correct answer from the options given below :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
- (C) (A) is true but (R) is false
- (D) (A) is false but (R) is true

Q96 Of the following, which is NOT a logical error?

- (A)

Using the '=' instead of '===' to determine if two values are equal

- (B) Divide by zero
- (C) Failing to initialize counter and total variables before the body of loop
- (D) Using commas instead of two required semicolon in a for loop header

Q97 Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A) : A load-and-go assembler avoids the overhead of writing the object program out and reading it back in.

Reason (R) : This can be done with either one-pass or two pass assembler.

In the light of the above statements, choose the correct answer from the options given below :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
- (C) (A) is true but (R) is false
- (D) (A) is false but (R) is true

Q98 Which statement is false?

- (A) All function calls in C pass arguments using call by value.
- (B) Call by reference enables a called function to modify a variable in calling function.
- (C) Call by value is always more efficient than call by reference.
- (D) Programmers use pointers and indirection operation to simulate call by reference.

Q99 Given below are two statements :

Statement I: "Grandparent is a parent of one's parent".

Statement II : First Order Predicate Logic (FOPL) representation of above statement is

$$\forall g, c \text{ grandparent}(g, c) \Leftrightarrow \exists p \text{ parent}(g, p) \wedge \text{parent}(p, c)$$

In the light of the above statements, choose the most appropriate answer from the options given below :

- (A) Both Statement I and Statement II are correct
- (B) Both Statement I and Statement II are incorrect
- (C) Statement I is correct but Statement II is incorrect
- (D) Statement I is incorrect but Statement II is correct

Q100 Given below are two statements :

Statement I: Consider 20 bit 'Branch'

microinstruction code format given below:



3	3	3	2	2	7
F1	F2	F3	CD	BR	AD

F1, F2, F3 : Micro-operation fields

CD : Condition for branching

BR : Branch field

AD : Address field

Statement II: Instruction represented in above format can perform branch in 4 conditions.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (A) Both Statement I and Statement II are correct
- (B) Both Statement I and Statement II are incorrect
- (C) Statement I is correct but Statement II is incorrect
- (D) Statement I is incorrect but Statement II is correct



CUSTOMER
SERVICE

Answer Key

Q1 (B)
Q2 (A)
Q3 (A)
Q4 (A)
Q5 (A)
Q6 (D)
Q7 (D)
Q8 (D)
Q9 (D)
Q10 (A)
Q11 (C)
Q12 (A)
Q13 (A)
Q14 (C)
Q15 (A)
Q16 (A)
Q17 (A)
Q18 (C)
Q19 (D)
Q20 (B)
Q21 (D)
Q22 (B)
Q23 (B)
Q24 (C)
Q25 (A)
Q26 (A)
Q27 (D)
Q28 (C)
Q29 (D)
Q30 (D)
Q31 (A)
Q32 (A)
Q33 (B)
Q34 (B)
Q35 (A)
Q36 (A)
Q37 (B)
Q38 (B)
Q39 (A)

Q40 (C)
Q41 (D)
Q42 (A)
Q43 (D)
Q44 (D)
Q45 (B)
Q46 (B)
Q47 (C)
Q48 (B)
Q49 (B)
Q50 (B)
Q51 (A)
Q52 (C)
Q53 (B)
Q54 (A)
Q55 (B)
Q56 (C)
Q57 (B)
Q58 (D)
Q59 (C)
Q60 (D)
Q61 (A)
Q62 (D)
Q63 (C)
Q64 (A)
Q65 (A)
Q66 (D)
Q67 (A)
Q68 (D)
Q69 (A)
Q70 (A)
Q71 (C)
Q72 (C)
Q73 (D)
Q74 (A)
Q75 (D)
Q76 (C)
Q77 (C)
Q78 (B)



Q79 (D)
Q80 (C)
Q81 (C)
Q82 (B)
Q83 (B)
Q84 (D)
Q85 (C)
Q86 (A)
Q87 (A)
Q88 (B)
Q89 (D)

Q90 (C)
Q91 (C)
Q92 (A)
Q93 (C)
Q94 (D)
Q95 (B)
Q96 (D)
Q97 (A)
Q98 (C)
Q99 (A)
Q100 (A)



Hints & Solutions

Q1 Text Solution:

Option B is correct

To determine the certainty factor (CF) of the result based on the premises provided, let's break down the given information

- Rule: (P1 and P2) or P3 $\Rightarrow$ R1(0.8) and R2(0.3)
- Premises with certainty factors:
 - CF(P1)=0.5
 - CF(P2)=0.8
 - CF(P3)=0.2

Step 1: Compute CF for (P1 and P2)

Using the **minimum rule** for "AND":

$$CF(P1 \text{ and } P2) = \min(CF(P1), CF(P2)) = \min(0.5, 0.8) = 0.5$$

Step 2: Compute CF for (P1 and P2) or P3

Using the **maximum rule** for "OR":

$$CF((P1 \text{ and } P2) \text{ or } P3) = \max(CF(P1 \text{ and } P2), CF(P3)) = \max(0.5, 0.2) = 0.5$$

Step 3: Apply CFs of Conclusions (R1 and R2)

The rule states that:

- (P1 and P2) or P3 $\Rightarrow$ R1 with CF=0.8
- (P1 and P2) or P3 $\Rightarrow$ R2 with CF=0.3

The **resulting CF for R1** is calculated by multiplying the CF of (P1 and P2) or P3 with the CF of R1:

$$CF(R1) = CF((P1 \text{ and } P2) \text{ or } P3) \times CF(R1) = 0.5 \times 0.8 = 0.4$$

The **resulting CF for R2** is calculated similarly:

$$CF(R2) = CF((P1 \text{ and } P2) \text{ or } P3) \times CF(R2) = 0.5 \times 0.3 = 0.15$$

Hence, CF(R1) = 0.4 and CF(R2) = 0.15

Q2 Text Solution:

Option A is correct

B1, B2, B3: Events of choosing the first, second, or third box, respectively. Since the box is chosen at random:

$$P(B1) = P(B2) = P(B3) = 1/3$$

E: Event of drawing 1 white and 1 red ball from the selected box.

Now, Calculate $P(E|B1)$ for each Box as below:

For Box B1,

Total balls: 9

$$\text{Total ways to choose 2 balls: } \frac{9!}{2! \times (9-2)!} = 36$$

(Binomial Coefficient)

Ways to choose 1 white and 1 red ball:

- White balls: $= \frac{2!}{1! \times (2-1)!} = 2$ ways
- Red balls: $= \frac{4!}{(1!) \times (4-1)!} = 4$ ways
- Total ways: $2 \times 4 = 8$

Hence, Probability is $P(E|B1) = 8/36 = 2/9$

Similarly for Box B2,

Total Balls: 7

$$\text{Total ways to choose 2 balls: } \frac{7!}{2! \times (7-2)!} = 21$$

(Binomial Coefficient)

Ways to choose 1 white and 1 red ball:

- White balls $= \frac{3!}{1! \times (3-1)!} = 3$ ways
- Red balls $= \frac{2!}{1! \times (2-1)!} = 2$ ways
- Total ways: $3 \times 2 = 6$

Hence, Probability is $P(E|B2) = 6/21 = 2/7$

Similarly for Box B3,

Total Balls: 8

$$\text{Total ways to choose 2 balls} = \frac{8!}{(2!) \times (8-2)!} = 28 \text{ (Binomial Coefficient)}$$

Ways to choose 1 white and 1 red ball:

- White balls = 4 ways
- Red balls = 3 ways
- Total ways: $4 \times 3 = 12$

Hence, Probability is $P(E|B2) = 12/28 = 3/7$

Using Law of Total Probability,

$$\text{Total Probability} = P(E) = P(E|B1) \times P(B1) + P(E|B2) \times P(B2) + P(E|B3) \times P(B3)$$

$$= (2/9) \times (1/3) + (2/7) \times (1/3) + (3/7) \times (1/3)$$

$$P(E) = 59/189$$

Now Applying Bayes Theorem (for finding $P(B1|E)$),

$$P(B1|E) = \{P(E) \times P(E|B1)\} / P(B1) = (2/9 \times 1/3) /$$

$$(59/189) = 14/59 = 0.237$$

Q3 Text Solution:

Option A is correct

For $m \ll n$: The access time is reduced due to interleaving, as each bank can handle requests independently. Therefore, the access time depends on m as $m \cdot T/n$, benefiting from interleaving across banks.

For $m \gg n$: With larger m , requests frequently collide in the same banks, and each access requires the full bank cycle time T , as interleaving offers no advantage.

Q4 Text Solution:

Option A is Correct.

A **crossbar switch** in a multiprocessor system is an example of a **non-blocking network** because it allows any processor to access any memory module or I/O device without interfering with other connections. Each processor-memory pair has a dedicated path within the crossbar, so multiple connections can occur simultaneously without blocking one another.

In contrast, a **blocking network** would mean that certain connections prevent others from



occurring at the same time, which is not the case for crossbar switches.

Q5 Text Solution:

Option A is Correct

According to the question, Hamming Code Structure is (7,4)

In a (7,4) Hamming code:

- 4 data bits (D1, D2, D3, D4) are combined with 3 parity bits (P1, P2, P4).
- The parity bits are placed at positions that are powers of 2 (1, 2, and 4).
- The data bits are placed in the remaining positions (3, 5, 6, and 7).

Step 1: Assign Data Bits to Positions

Given data bits 1101, we assign them as follows:

- D1 = 1 goes to position 3.
- D2 = 1 goes to position 5.
- D3 = 0 goes to position 6.
- D4 = 1 goes to position 7.

Our initial layout (with placeholders for parity bits) is:

Position	1	2	3	4	5	6	7
	(P1)	(P2)	(D1)	(P4)	(D2)	(D3)	(D4)
Bits	?	?	1	?	1	0	1

Step 2: Calculate Parity Bits to Ensure Even Parity
Parity Bit P1 (Position 1)

- Covers positions: 1, 3, 5, and 7.
- Relevant bits in these positions (excluding P1 itself) are: 1 (D1), 1 (D2), and 1 (D4).
- Sum of these bits = $1+1+1=3$ (odd).
- To make this even, P1 should be 1.

Parity Bit P2 (Position 2)

- Covers positions: 2, 3, 6, and 7.
- Relevant bits in these positions (excluding P2 itself) are: 1 (D1), 0 (D3), and 1 (D4).
- Sum of these bits = $1+0+1=2$ (even).
- Since the sum is already even, P2 should be 0.

Parity Bit P4 (Position 4)

- Covers positions: 4, 5, 6, and 7.
- Relevant bits in these positions (excluding P4 itself) are: 1 (D2), 0 (D3), and 1 (D4).
- Sum of these bits = $1+0+1=2$ (even).
- Since the sum is already even, P4 should be 0.

Step 3: Construct the Final 7-Bit Hamming Code
With the parity bits calculated, the final 7-bit Hamming code is:

Position	1	2	3	4	5	6	7
	(P1)	(P2)	(D1)	(P4)	(D2)	(D3)	(D4)
Bits	1	0	1	0	1	0	1

So, the final 7-bit Hamming code with even parity for the data 1101 is: 1010101

Q6 Text Solution:

Option D is correct

In digital logic design problems, the total number of gate inputs is calculated by:

- Counting the number of literals (variables and their complements).
- Adding the number of terms (inputs to the OR gate).

Counting the Literals,

- First Term ABC is having 3 Literals,
- Second Term is having 4 literals,
- Third Term is having 2 literals, and
- Fourth Term AD is having 2 literals. Hence,
Total number of Literals = $3+4+2+2 = 11$

Counting the Inputs for OR Gate: 4.

The Total Number of Gates are $11+4 = 15$.

The discrepancy arises due to different conventions in counting gate inputs:

- Detailed Counting (17 inputs): Includes all gate inputs (NOT, AND, and OR gates).
- Standard Counting in Exam Problems (15 inputs): Counts the number of literals and the number of terms (inputs to the OR gate).

Q7 Text Solution:

Option D is correct

The circuit has:

- 8 input lines: $D_0, D_1, \dots, D_7$
- 3 output lines: A_2, A_1, A_0

The outputs are defined by:

1. $A_2 = D_4 + D_5 + D_6 + D_7$
2. $A_1 = D_2 + D_3 + D_6 + D_7$
3. $A_0 = D_1 + D_3 + D_5 + D_0$

Here, the "+" symbol represents the logical OR operation.

Let's break down each output equation to understand the functionality:

1. Output A_2 :

- Defined as $A_2 = D_4 + D_5 + D_6 + D_7$.
- This output line is set to 1 if any of the inputs D_4, D_5, D_6 , or D_7 is 1.

2. Output A_1 :

- Defined as $A_1 = D_2 + D_3 + D_6 + D_7$.
- This output line is set to 1 if any of the inputs D_2, D_3, D_6 , or D_7 is 1.

3. Output A_0 :

- Defined as $A_0 = D_1 + D_3 + D_5 + D_0$.



- This output line is set to 1 if any of the inputs D_1 , D_3 , D_5 , or D_0 is 1.

The purpose of this circuit is to produce a specific output pattern based on combinations of inputs, but it does not identify a single input line or prioritize inputs as a priority encoder would. As by definition, encodes multiple active inputs into a binary code representing the highest-priority active input. This problem involves understanding how multiple inputs are encoded into fewer outputs based on specific priority rules. Priority encoders are fundamental components in digital systems for tasks like interrupt handling, where the system needs to prioritize one signal over others. Hence, the circuit behaves like a Priority Encoder, encoding the highest-priority active input into a binary code.

Q8 Text Solution:

Option D is correct
Given Information

1. Number of tracks on the floppy disk: 80
2. Bytes per sector: 128 bytes
3. Total storage capacity: 163,840 bytes
4. Unknown: Number of sectors per track (denoted as S)

Set up the equation for total storage capacity:
Total storage capacity = Number of tracks $\times$
Number of sectors per track $\times$ Bytes per sector
Substituting the values:
 $163,840 = 80 \times S \times 128$
 $S = 163,840 / (80 \times 128)$
 $S = 16$

Q9 Text Solution:

Option D is Correct
Given Information

1. Tag field: 9 bits
2. Index field: 12 bits

The total address size in bits is:
Total address bits = $9 + 12 = 21$ bits

- Step 1: Calculate the Size of the Main Memory

The size of the main memory is determined by the total number of addressable locations, which is $2^{\text{total address bits}}$.

With 21 address bits:
Main memory size = 2^{21} bytes = 2,048 KB
or equivalently, Main memory size = 2 MB

- Step 2: Calculate the Size of the Cache Memory

The Index field represents the number of cache lines. Since there are 12 bits in the Index field, the number of cache lines is:

Number of cache lines = $2^{12} = 4096$

Assuming each cache line represents one byte of storage (since no additional Offset bits are specified):

Cache memory size = 4096 bytes = 4 KB

Hence, the answer is 2048K, 4K

Q10 Text Solution:

The primal problem is:

Maximize $z = 5x_1 + 12x_2 + 4x_3$

Subject to:

$$x_1 + 2x_2 + x_3 = 10$$

$$2x_1 - x_2 + 3x_3 = 8$$

$$x_1, x_2, x_3 \geq 0$$

Dual Problem Formulation

For a maximization primal problem with equality constraints, the dual problem will be a minimization problem with inequality constraints.

1. Objective Function of the Dual:

- The constants on the right-hand side of the primal constraints (10 and 8) become the objective coefficients in the dual.
- Thus, the dual objective function is:

Minimize $w = 10y_1 + 8y_2$

1. Dual Constraints:

- Each variable in the primal (e.g., x_1 , x_2 , x_3) gives rise to a constraint in the dual.
- The coefficients of each variable in the primal constraints become the coefficients in the dual constraints.

So, the dual constraints are:

- For x_1 : $y_1 + 2y_2 \geq 5$
- For x_2 : $2y_1 - y_2 \geq 12$
- For x_3 : $y_1 + 3y_2 \geq 4$

1. Sign Restrictions on Dual Variables:

- Since the primal constraints are equalities, the dual variables y_1 and y_2 are unrestricted in sign.

where both y_1 and y_2 are unrestricted in sign.

In our primal problem:

1. Primal Problem:

- The primal problem has equality constraints:

$$x_1 + 2x_2 + x_3 = 10$$

$$2x_1 - x_2 + 3x_3 = 8$$

- Each constraint is an equality, not an inequality.

1. Dual Variables:

- According to the general rules stated, if the primal constraints are equality constraints,



the corresponding dual variables y_1 and y_2 should be unrestricted in sign.

Both y_1 and y_2 are unrestricted in sign in the dual problem.

If required to choose the closest answer, Option A (" $y_1 \geq 0$, y_2 unrestricted") is the least incorrect, but ideally, both variables should be unrestricted in sign.

Q11 Text Solution:

Option C is Correct

Constructing the Truth Table for this Expression

P	Q	$(P \vee Q)$	$(\overline{P} \wedge \overline{Q})$	$(P \wedge \overline{Q})$	$(\overline{P} \wedge Q)$	$(\overline{P} \wedge Q) \vee (P \wedge \overline{Q})$
0	0	0	0	0	0	0
0	1	1	1	0	0	1
1	0	1	0	1	0	1
1	1	1	0	0	1	1

From the truth table, we observe that the expression is true in all cases except when both P and Q are false. The resulting truth table matches the truth table for the OR operation between P and Q, which is true in all cases except when both P and Q are false.

Thus, the expression is equivalent to $P \vee Q$

Q12 Text Solution:

To reduce the grammar, we need to identify any **non-productive symbols** (symbols that cannot derive a terminal string) and **unreachable symbols** (symbols that cannot be reached from the start symbol S).

1. Identify Non-Productive Symbols:

- B has the productions $B \rightarrow BC \mid AB$, both of which involve B itself and do not lead to a terminal string directly.
- This makes B **non-productive**, as it cannot derive any terminal string.

2. Removing Non-Productive Symbols:

- Since B is non-productive, we can remove all productions involving B, including $S \rightarrow AB$, $B \rightarrow BC$, $B \rightarrow AB$, and any rules in C involving B.

$S \rightarrow AB \mid CA$

$B \rightarrow BC \mid AB$

$A \rightarrow a$

$C \rightarrow aB \mid b$

Here B is not derive any terminal string so remove productions contains B $S \rightarrow CA$ $A \rightarrow a$ $C \rightarrow b$
After removing these, the grammar simplifies to final grammar which is option A:

$S \rightarrow CA$

$A \rightarrow a$

$C \rightarrow b$

Q13 Text Solution:

Option A is Correct

The production rules are:

- $S \rightarrow AbB$
- $A \rightarrow aAb \mid \lambda$
- $B \rightarrow bB \mid \lambda$

Step-by-Step Analysis

Let's look at each non-terminal's role in generating strings.

1. Start Symbol S:

- The production rule for S is: $S \rightarrow AbB$
- This means any string generated by S will have the structure: (string from A)+b+(string from B)

2. Productions for A:

- $A \rightarrow aAb$: This rule allows A to produce a series of a's followed by b's, with each a paired with a b.
- $A \rightarrow \lambda$: This rule allows A to be empty, giving flexibility in how many pairs of a's and b's are generated.
- Therefore, A can generate strings of the form $a^n b^n$, where $n \geq 0$.

3. Productions for B:

- $B \rightarrow bB$: This rule allows B to generate additional b's.
- $B \rightarrow \lambda$: This rule allows B to be empty if desired.
- So, B can generate strings of the form b^m , where $m \geq 0$.

Hence, putting all together **Putting It All Together**

Given $S \rightarrow AbB$, analyse the overall structure:

- The part A generates $a^n b^n$, with equal numbers of a's and b's.
- The middle b from $S \rightarrow AbB$ adds an additional b.
- B generates b^m , allowing for any additional number of b's.

Thus, any string generated by S has the structure:

$a^n b^{n+1+m}$

where $n \geq 0$ and $m \geq 0$.

So, the language L generated by G is:



$$L = \{a^n b^k \mid n \geq 0, k \geq n+1\}$$

In other words:

- We have n a's followed by k b's.
- The number of b's (k) is strictly greater than the number of a's (n).

Option A, states that the number of b's (m) is strictly greater than the number of a's (n), which aligns with the structure $a^n b^{n+1+m}$ where $m \geq 0$.

This matches our derived language L.

Q14 Text Solution:

Option C is Correct

The language $L = \{a^n b^m : n \geq 4, m \leq 3\}$ specifies that:

1. **$n \geq 4$:** Each string must contain at least four a's.
 - This can be represented by starting with exactly four a's (i.e., aaaa), followed by any number of additional a's, represented by a^* .
2. **$m \leq 3$:** Each string can contain 0, 1, 2, or 3 b's.
 - This can be represented by $\lambda + b + bb + bbb$, where λ represents the option for no b's.

For given Language L, the regular expression should satisfy the following conditions:

1. At least 4 a's:

- The expression must start with exactly four a's, ensuring the minimum condition for $n \geq 4$.
- This can be represented as aaaa.

2. Additional a's allowed:

- We use a^* after the initial four a's to allow for any additional a's if $n > 4$.
- So, aaaaa* covers this requirement.

3. At most 3 b's:

- The expression should allow between 0 and 3 b's.
- This can be represented as $(\lambda + b + bb + bbb)$.

4. Order of a's and b's:

- The b's should come after the a's, so the regular expression must keep the b-portion after the a-portion.

The expression that satisfies all these conditions is:

$$aaaaa^*(\lambda + b + bb + bbb)$$

Q15 Text Solution:

Option A is correct

The language $L = \{ab, aa, baa\}$ consists of three strings:

1. ab
2. aa
3. baa

The set L^* represents the **Kleene closure** of L , which means it includes all strings that can be formed by concatenating zero or more elements from L in any order. Therefore, any string in L^* must be composed of repetitions of the strings ab, aa, and baa.

Option A: baaaaabaaaa

Let's try to partition baaaaabaaaa into valid substrings from L :

1. Starting from the beginning, we see the sequence "baa aa ab aa aa a".
 - This does not match any of the strings in L and cannot be partitioned to match any combination of ab, aa, or baa.

Conclusion: Option A is NOT in L^* .

Option B: abaabaaabaa

Let's try to partition abaabaaabaa:

1. We can partition it as follows:
 - "abaabaaabaa"
2. Each of these substrings:
 - ab, aa, baa, ab, and aa are all elements of L .

Conclusion: Option B is in L^* .

Option C: aaaabaaaa

Let's try to partition aaaabaaaa:

1. We can partition it as follows:
 - "aaaabaa"
2. Each of these substrings:
 - aa, aa, ab, and aa are all elements of L .

Conclusion: Option C is in L^* .

Option D: baaaaabaa

Let's try to partition baaaaabaa:

1. We can partition it as follows:
 - "baaaaab aa"
2. Each of these substrings:
 - baa, aa, and baa are all elements of L .

Conclusion: Option D is in L^* .

Hence, **Option A: baaaaabaaaa is NOT in L^* .**

Options B, C, and D are all in L^* , as they can be fully partitioned into substrings that belong to L .

Q16 Text Solution:

Option A is Correct

The NPDA is defined as:

- States: $Q = \{q_0, q_1, q_f\}$
- Input Alphabet: $\Sigma = \{a, b\}$
- Stack Alphabet: $\Gamma = \{1, z\}$
- Start State: q_0
- Start Stack Symbol: z
- Accepting State: q_f



- Transition Function δ :
 - $\delta(q_0, \lambda, z) = \{(q_f, z)\}$: Allows an immediate transition to the accepting state q_f with the initial stack symbol z , which means the NPDA accepts the empty string.
 - $\delta(q_0, a, z) = \{(q_1, 11z)\}$: On input a and stack top z , push two 1's onto the stack and transition to q_1 .
 - $\delta(q_1, a, 1) = \{(q_1, 111)\}$: On input a and stack top 1, push two more 1's onto the stack (effectively pushing three 1's for each a in q_1).
 - $\delta(q_1, b, 1) = \{(q_1, \lambda)\}$: On input b and stack top 1, pop the stack.
 - $\delta(q_1, \lambda, z) = \{(q_f, z)\}$: If the stack has returned to z in q_1 with no remaining input, move to the accepting state q_f .

Step-by-Step Analysis of the Language

1. Processing a's in State q_1 :

- When the NPDA reads an a in state q_0 with z on the stack, it transitions to q_1 and pushes two 1's onto the stack.
- For each subsequent a read in q_1 with 1 on the stack, the NPDA pushes two additional 1's onto the stack.
- Therefore, for every a in the input, the stack gains two 1's.

This setup implies that after reading $2n$ a's, the stack will have $n \times 2 = 2n$ 1's.

1. Processing b's in State q_1 :

- When the NPDA reads a b in q_1 with 1 on top of the stack, it pops the stack.
- Therefore, each b in the input corresponds to one pop operation on the stack.

2. Balancing the a's and b's:

- For the NPDA to reach the accepting state q_f (while consuming all input), the number of b's must match the number of pops required to empty the stack of the 1's pushed by the a's.
- Since each a contributes two 1's to the stack, there must be twice as many b's as a's in any accepted string.
- Therefore, the language accepted by the NPDA is: $L = \{a^{2n}b^n \mid n \geq 0\}$

drawing them from farthest to nearest, so that closer objects overwrite farther ones.

This algorithm is indeed a solution for the hidden surface removal problem, especially in simpler graphics pipelines. While it may not be as efficient as Z-buffering for complex scenes, it is effective for minimal 3D pipelines where resources are limited.

While brute force rasterization can solve the hidden surface removal problem by calculating visibility at each pixel, it is generally computationally intensive and not as efficient as other methods. For simple 3D graphics pipelines or small scenes, it may be usable, but in practice, more optimized algorithms are usually preferred. The Painter's Algorithm is a specific solution for hidden surface removal in 3D rendering. Brute force rasterization does not solve this problem, as it lacks depth management and visibility checking, leading to potentially incorrect results in a 3D scene.

Q18 Text Solution:

first we will calculate the value of n

$$n = p * q$$

$$n = 13 * 5$$

$$\text{so } n = 65$$

next calculate $\phi(n)$:

$$\phi(n) = (p-1) * (q-1)$$

$$(13-1) * (5-1)$$

$$\text{so } \phi(n) = 48$$

now, we need to find the modular multiplicative inverse of 'e'

modul $\phi(n)$. In other words, we need to find 'd' such that $(e * d) \bmod \phi(n) = 1$.

Using the extended Euclidean algo, we can find 'd'.

$$\text{step 1: } 48 = 7 * 6 + 6$$

$$7 = 6 * 1 + 1$$

$$\text{Step 2: } 6 = 1 * 6 + 0$$

since the remainder in the last step is 1, we can conclude that the greatest common divisor of 7 and 48 is 1. Therefore, 'd' exists, and it is the coefficient of 7 in the equation:

$$1 = 7 - 6 * 1$$

$$\text{so } d = 7$$

$$\text{now plaintext } y = (6^7) \bmod 65$$

$$\text{plaintext } 279936 \% 65$$

$$\text{plaintext } = 46$$

$$\text{so the } d = 7$$

and the plaintext of the ciphertext '6' will be 46.

Q17 Text Solution:

Option A is Correct

The Painter's algorithm is a commonly used method for hidden surface removal. It works by sorting polygons by their depth and then

Q19 Text Solution:



The expression $!(num=65)$ evaluates to true if num is not equal to 65, and false if num is equal to 65.

Q20 Text Solution:

Correct option is B.

A pointer is a variable whose value is the address of another variable, i.e., that direct address of the memory location. Like any variable or constant, you must declare a pointer before using it to store any variable address.

Q21 Text Solution:

A namespace allows organizations to specify globally unique names as element tags in documents, as each element name can be associated with a different namespace.

Q22 Text Solution:

In systems that use cascading termination, all child process are terminated if the parent process terminates, whether normally or abnormally.

Q23 Text Solution:

Correct option is B.

Using a transaction log, or journal, journaling improves the file system's dependability and enables rapid crash recovery. The journal is an on-disk log of metadata, or data about the file system, which is kept current as the file system is modified. Without journaling, file systems save changes to update in memory.

Q24 Text Solution:

Correct option is C.

Homography, also referred to as planar homography, is a transformation that is occurring between two planes. In other words, it is a mapping between two planar projections of an image. It is represented by a 3×3 transformation matrix in a homogenous coordinates space.

Q25 Text Solution:

The RAD model is a type of software development methodology that emphasizes quick and iterative releases cycles, primarily focusing on delivering working software in shorter timeliness.

Q26 Text Solution:

Verifiable - an SRS is verifiable only if every stated requirement can be verified cost effectively.

Q27 Text Solution:

Mutation Testing Fault-based testing strategy. Mutation Testing Faults are introduced into the program by creating versions of the program called mutants.

Q28 Text Solution:

Correct option is C.

Acceptance testing guarantees that software aligns with user needs and business requirements. The major aim of this test is to evaluate the compliances of the system with the business requirements and assess whether it is acceptable for delivery or not.

Alpha and Beta testing comes under the Acceptance testing. Alpha testing is used to determine the product in the development testing environment by a specialized testers team while Beta testing is used to assess the product by exposing it to the real end users.

Q29 Text Solution:

The process of gathering requirements from clients and analyze and document them is called requirement engineering.

Q30 Text Solution:

Product Metrics describes the characteristics of the product such as size, complexity, design features, performance, and quality level.

Q31. Text Solution:

The correct answer is **option A**.

Query: $\pi_{sid}(\delta_{age > 20} Sailors) - \pi_{sid}((\delta_{color = red} Boats) \bowtie Reserves \bowtie \Delta Sailors)$

$\delta_{age > 20} Sailors$: This part selects sailors older than 20.

$\pi_{sid}((\delta_{color = red} Boats) \bowtie Reserves \bowtie \Delta Sailors)$:

This part finds SIDs of sailors who have reserved red boats.

The final operation is a set difference between these two results.

This option correctly identifies sailors over age 20 and excludes those who have reserved red boats.

Q32. Text Solution:

The correct answer is **option A**.

Valid: Q1

$\pi_{sname}((\delta_{color = red} Boats) \bowtie Reserves \bowtie Sailors)$

This query attempts to select red boats from the Boats table, then joins with the Reserves and Sailors tables. Thus, this query is valid.

Valid: Q2 $\pi_{sname}(\pi_{sid}((\pi_{bid}$

$\delta_{color = red} Boats) \bowtie Reserves) \bowtie Sailors)$

This query correctly selects the bids of red boats from the Boats table, joins with Reserves to get the corresponding sailor IDs, and then joins with Sailors to get the names. This query is valid and should return the names of sailors who reserved red boats.

Invalid: Q3 $\pi_{sname}(\delta_{color = red} Reserves) Boats \bowtie Sailors)$



This query appears to be trying to filter Reserves instead of Boats for red color, which is not the correct approach. Thus, this query is also invalid.

Q33. Text Solution:

The correct answer is **option B**.

Incorrect : Q1 $\pi_{\text{name}}((\delta_{\text{bid}=103}\text{Boats})\bowtie\text{Sailors})$

This query attempts to select boat 103 from the Boats table and then join it with the Sailors table. However, this is incorrect because it does not involve the Reserves table, which is necessary to find which sailors reserved that boat. Thus, this query is invalid.

Correct : Q2 $\pi_{\text{name}}((\delta_{\text{bid}=103}(\text{Reserves}\bowtie\text{Sailors}))$

This query attempts to join the Reserves table with the Sailors table first and then select rows where the boat ID is 103. Thus, this query is valid.

Correct : Q3 $\pi_{\text{name}}((\delta_{\text{bid}=103}\text{Reserves})\bowtie\text{Sailors})$

This query correctly selects the rows from the Reserves table where the boat ID is 103 and then joins the result with the Sailors table to get the names of those sailors.

Q34. Text Solution:

The correct answer is **option B**.

$B:\rho(\text{Tempsids}, (\pi_{\text{sid}, \text{bid}}\text{Reserves}) / \pi_{\text{bid}}\text{Boats}) \pi_{\text{name}}((\text{Tempsids})\bowtie\text{Sailors})$

This is trying to find sailors who have reserved all boats by using a division operation. The result of this division would give the sailor IDs of those who reserved all boats. Then it joins with the Sailors table to get the names. This is valid.

Q35. Text Solution:

The correct answer is **option A**.

$A:\rho(\text{Tempred}, \pi_{\text{sid}}((\delta_{\text{color}=\text{Boats' red'}})\bowtie\text{Reserves})), \rho(\text{Tempgreen}, \pi_{\text{sid}}((\delta_{\text{color}=\text{Boats' green'}})\bowtie\text{Reserves})), \pi_{\text{name}}(\text{Tempred} \cap \text{Tempgreen}) \bowtie \text{Sailors}$

This option correctly identifies sailors who reserved red boats and those who reserved green boats, then finds the intersection of these two sets of sailor IDs. Finally, it retrieves the names of those sailors. This option is valid.

Q36. Text Solution:

The correct answer is **option A**.

Transmission Delay (T_t): The transmission delay is calculated using the formula:

$$T_t = \frac{\text{Frame Size}}{\text{Transmission Rate}}$$

Frame Size = 64 bytes = 64 * 8 bits = 512 bits

Transmission Rate = 1.536 Mbps = 1.536 * 10⁶ bits per second

Now, substituting the values into the formula:

$$T_t = \frac{512 \text{ bits}}{1.536 \times 10^6 \text{ bits/sec}} = 333.33 \text{ } \mu\text{sec}$$

Propagation Delay (T_p):

Distance = 3000 km

Propagation Speed = 6 $\mu\text{sec/km}$

For 1 km it takes 6 μsec ,

For 3000 km it will take 3000 * 6 μsec = 18000 μsec .

Q37. Text Solution:

The correct answer is **option B**.

The minimum number of bits required in the packet's sequence number field in the sliding window protocol is calculated using the formula = $\text{ceil}(1 + 2a)$, where $a = \frac{T_p}{T_t}$.

$T_p = 18000 \text{ } \mu\text{sec}$ $T_t = 333.33 \text{ } \mu\text{sec}$

Minimum number of bits in sequence number = $\text{ceil}[\log_2(1 + 2a)]$

$$1 + 2a = 1 + \frac{2T_p}{T_t} = 1 + \frac{2 \times 18000}{333.33} = 109.001$$

$$\text{ceil}[\log_2(1 + 2a)] = \text{ceil}[\log_2(109.001)] = 7$$

Q38. Text Solution:

The correct answer is **option B**.

To determine the sender window size that maximizes efficiency in a sliding window protocol, we can use the following formula:

Window Size = $1 + 2a$ where $a = T_p/T_t$

$T_p = 18000 \text{ } \mu\text{sec}$ $T_t = 333.33 \text{ } \mu\text{sec}$

window size = $1 +$

$$\frac{2T_p}{T_t} = 1 + \frac{2 \times 18000}{333.33} = 1 + 108.001 = 109.001$$

This means the sender can have up to 109 frames in transit before needing an acknowledgment, maximizing the utilization of the available bandwidth in the trunk.

Q39. Text Solution:

The correct answer is **option A**.

The transmission delay (T_t) = 333.33 μsec .

The propagation Delay (T_p) = 18 000 μsec .

Determine Window Size : With 6 bits reserved for the sequence number field, the maximum window size (N) can be calculated as: $N = 2^6 = 64$

$1 + 2a = 1 +$

$$\frac{2 \times T_p}{T_t} = 1 + \frac{2 \times 18000}{333.33} = 1 + 108.001 \approx 109$$

Calculate Efficiency : Efficiency (E) of the sliding window protocol can be calculated using the formula:

$$E = \frac{N}{1 + 2a} \text{ where } a = T_p/T_t$$

$$E = 64/109 \approx 0.587$$

Q40. Text Solution:

The correct answer is **option C**.

The maximum achievable throughput = efficiency (E) x bandwidth (B)

where E = 0.587 and B = 1.536 mbps

The maximum achievable throughput = E x B = 0.587 x 1.536 mbps = 0.901 (approx)

Q41 Text Solution:



Size-oriented metrics have been derived by normalizing quality and productivity is measured by considering the size of the software produced. The organization builds a simple record of size measures for the software projects.

Q42 Text Solution:

The evolutionary development approach is where requirements are implemented in an evolving, iterative manner, often categorized and refined in each iteration.

Q43 Text Solution:

All of these are correct
Software measurement has two types : first one is direct measure and second is indirect measure. In indirect measure, it include products like functionality, quality, complexity, reliability, maintainability and many more.

Q44 Text Solution:

Option D is correct
When two elements interact with the same input or proudce the same output then this is called communicational cohesion.

Q45 Text Solution:

- Cipher Feedback (CFB) Mode enables a block cipher to function as a self-synchronizing stream cipher.
- In CFB mode, the previous ciphertext block is encrypted rather than directly encrypting the plaintext.
- The encrypted output of the previous block is then XORed with the current plaintext block to generate the new ciphertext block.
- This process allows CFB mode to behave similarly to a stream cipher, with the added benefit of self-synchronization.

Q46 Text Solution:

- UDP is the connectionless transport-layer protocol used for applications requiring fast, efficient, but not necessarily reliable data transmission.

Q47 Text Solution:

Given information :

- Channel bandwidth: 64 kbps (kilobits per second)
- Frame size: 512 bytes
- Acknowledgments: Very short, we can ignore the time spent on acknowledgments.
- Window size: 15

Step 1: Convert Frame Size to Bits:

$$512 \text{ bytes} = 512 \times 8 = 4096 \text{ bits}$$

$$512 \text{ bytes} = 512 \times 8 = 4096 \text{ bits}$$

Step 2: Calculate the Transmission Time for One

Frame: Since the channel has a bandwidth of 64 kbps, the time to send one 4096-bit frame is:

Transmission time for one frame =

$$\frac{\text{Frame size in bits}}{\text{channel bandwidth}} = \frac{4096 \text{ bits}}{64000 \text{ bits/sec}} = 0.064 \text{ seconds} = 64 \text{ ms}$$

(ie, 1 second = 1000 milliseconds (ms) ---

$$0.064 \text{ seconds} = 0.064 \times 1000 = 64 \text{ milliseconds (ms))}$$

Step 3: Calculate the Window Throughput: The window size of 15 means we can send up to 15 frames before waiting for an acknowledgment.

Thus, the effective time we have for data transmission in one window is the RTT.

$$\text{Time for 15 frames} = 15 \times 0.064 \text{ seconds} = 0.96 \text{ seconds}$$

Step 4: Calculate the Total Bits Sent in One RTT:

In one RTT, we can send 15 frames, each with 4096 bits:

- Total bits in one RTT = $15 \times 4096 = 61440 \text{ bits}$

Step 5: Calculate Maximum Throughput:

Throughput is defined as the total bits sent divided by the RTT:

$$\text{Throughput} : 61440 \text{ bits} = 61.44 \text{ kbps (} 61.440 \text{ bits} \div 0.96 = 64 \text{ kbps) approx 64 kbps}$$

Final answer is :- 64 kbps.

Q48 Text Solution:

Given information: 167.199.170.82/27

Here, Number of Mask bits $n=27$

In classless addressing no of addresses are given by 2^{32-n} (Formula)

$$\text{No of addresses} = 2^{32-27} = 2^5 = 32$$

Final answer: 32 Addresses

Q49 Text Solution:

1. The Transport layer of the OSI model is responsible for dividing data into smaller segments or packets at the source and reassembling them at the destination.
2. It ensures reliable data transfer between devices by managing error correction, flow control, and retransmissions.
3. Protocols like TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) operate at this layer.

Q50 Text Solution:

Given Information according to the question:

Four stage pipeline is used Delay of stages = 150, 120, 160, and 140 ns

Delay due to each register = 5 ns

1000 data items or instructions are processed

Formula:-



1. Cycle Time = Cycle Time Maximum delay due to any stage + Delay due to its register
 2. Pipeline time to process 1000 data items = Time taken for 1st data item + Time taken for remaining 9 data items

Step1: Cycle Time = Cycle Time Maximum delay due to any stage + Delay due to its register

= Max {150, 120, 160, 140} + 5 ns

= 160 ns + 5 ns

= 165 ns

Step 2: Process:

Pipeline time to process 1000 data items

= Time taken for 1st data item + Time taken for remaining 9 data items

= 1 x 4 clock cycles + 999 x 1 clock cycle

= 4 x cycle time + 999 x cycle time

= 4 x 165 ns + 999 x 165 ns (Cycle time = 165 ns)

= 660 ns + 164835 ns

= 165495 ns

= 165.5 ms

Final answer: 165.5 ms.

Q51 Text Solution:

The given address, 4A:30:10:21:10:1A, is a unicast address. A unicast address refers to a unique, one-to-one communication between a single source and a single destination.

4 A can be written as 01001010 and its LSB bit is 0 so it is a unicast address.

Q52 Text Solution:

$f(n) = o(g(n))$ and $g(n) = o(h(n)) \Rightarrow f(n) = o(h(n))$

If $f(n)=o(g(n))$, then $f(n)$ grows strictly slower than $g(n)$, and if $g(n)=o(h(n))$, then $g(n)$ grows strictly slower than $h(n)$. From these, it logically follows that $f(n)=o(h(n))$, meaning $f(n)$ grows strictly slower than $h(n)$. This is a transitive property of little-o notation.

Q53 Text Solution:

Step 1: Compare with Master Theorem Format
 The Master Theorem applies to recurrence relations of the form:

$T(n) = 3T(n/4) + n \log n$ where $f(n) = n^{\log_b a}$ as $a=3$, $b=4$, $f(n) = n \log n$

step 2: Calculate $n^{\log_b a}$, we need to compute

$\log_b a$ which is equivalent of $\frac{\log a}{\log b}$

then putting value as $\frac{\log 3}{\log 4} = \frac{0.4771}{0.6021} \approx 0.793$

(approx)

Step 3:- compare $n^{\log_b a} \approx n^{0.793}$

Step 4: If $f(n) = \Omega(n^c)$ where $c > \log_b a$ then $T(n) = \Theta(f(n))$, $n \log n$ grows faster than $n^{0.793}$.

So $T(n) = \Theta(n \log n)$

Q54 Text Solution:

Explanation:

The binary Heap structure is a complete binary tree.

In a complete binary tree, at height h , the number of nodes is 2^h .

Example:

at $h=0$ (root level) = 2^0 node = 1 node,

at $h=1$ = 2^1 node = 2 node

at $h=2$ = 2^2 node = 4 node

at $h=3$ = 2^3 node = 8 node

therefore, 1,2,4,8,16,32.....

where n element heap, nodes of height is h .

The number of nodes at height h should limited to a fraction of n using nodes 1,2,4,8,16,32.....

using formula as $\frac{n}{2^{h+1}}$

The final answer is: A

Q55 Text Solution:

Minimum and Maximum Number of Keys per Node:

- Every node (except the root) in a B-tree must have at least $t-1$ keys.
- A node can have at most $2t-1$ key

If we use h for the height of the tree and n for the total number of nodes, we get an inequality that shows the minimum number of nodes, which is:

$$n \geq 1 + (t-1) \sum_{i=1}^h (2^{t^i-1}) = (2^{t^h} - 1)$$

whereas the height of h of the tree is $h \leq$

$$\log_t \frac{n+1}{2}$$

The final answer is: $h \leq \log_t \frac{n+1}{2}$

Q56 Text Solution:

The **Quick Sort** algorithm uses the **Divide and Conquer** approach to sort an array

These are follow these step solve problem:

Step 1: Divide: Select a pivot and partition the array around it, placing smaller elements to the left and larger elements to the right.

Step 2: Conquer: Recursively apply the same steps to the left and right sub-arrays.

Step 3: Combine: No explicit combination is needed; partitions are sorted in place.

Q57 Text Solution:

Analyze the Hash Table and Probing:

- Hash Table Size: 11
- Hash Function: $h(k) = k \bmod 11$
- Probing: Linear probing is used, meaning if a bin is already occupied, the next available bin is used (search sequentially).

Insert Each Key:

We insert the keys one by one and determine their positions in the hash table.

Key 43:



- $h(43)=43 \bmod 11=10$
- Bin 10 is empty $\rightarrow$ Insert 43 at index 10.

Key 36:

- $h(36)=36 \bmod 11=3$
- Bin 3 is empty $\rightarrow$ Insert 36 at index 3.

Key 92:

- $h(92)=92 \bmod 11=4$
- Bin 4 is empty $\rightarrow$ Insert 92 at index 4.

Key 87:

- $h(87)=87 \bmod 11=10$
- Bin 10 is occupied $\rightarrow$ Linear probing: Check bin 0.
- Bin 0 is empty $\rightarrow$ Insert 87 at index 0.

Key 11:

- $h(11)=11 \bmod 11=0$
- Bin 0 is occupied $\rightarrow$ Linear probing: Check bin 1.
- Bin 1 is empty $\rightarrow$ Insert 11 at index 1.

Key 47:

- $h(47)=47 \bmod 11=3$
- Bin 3 is occupied $\rightarrow$ Linear probing: Check bin 4, then 5.
- Bin 5 is empty $\rightarrow$ Insert 47 at index 5.

Key 11 (Duplicate):

- $h(11)=11 \bmod 11=0$
- Bin 0 is occupied $\rightarrow$ Linear probing: Check bins 1, 2.
- Bin 2 is empty $\rightarrow$ Insert 11 at index 2.

Key 13:

- $h(13)=13 \bmod 11=2$
- Bin 2 is occupied $\rightarrow$ Linear probing: Check bin 3, 4, 5, 6.
- Bin 6 is empty $\rightarrow$ Insert 13 at index 6.

Key 14:

- $h(14)=14 \bmod 11=3$
- Bin 3 is occupied $\rightarrow$ Linear probing: Check bin 4, 5, 6, 7.
- Bin 7 is empty $\rightarrow$ Insert 14 at index 7.

The last record (14) is inserted at bin 7.

Q58 Text Solution:

Explanation:

Preorder $\rightarrow$ ABCEIFJDGHKL

In order $\rightarrow$ EICFJBGDKHLA

Node A $\rightarrow$ Left child: (EICFJBGDKHL), Right child: No child

Node B $\rightarrow$ Left child: (EICFJ), Right child: GDKHL

Node C $\rightarrow$ Left child: (EI), Right child: (FJ)

Node D $\rightarrow$ Left child: (G), Right child: (KHL)

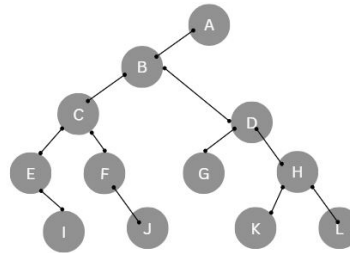
Node E $\rightarrow$ Left child: No child, Right child: (I).

Node F $\rightarrow$ Left child: No child, Right child: (J)

Node G $\rightarrow$ Left child: No child, Right child: No child

Node H $\rightarrow$ Left child: K, Right child: L

Post order: Left $\rightarrow$ Right $\rightarrow$ Node



The Final Answer is: IEJFCGKLHDBA

Q59 Text Solution:

Option C is Correct

To determine how many rotations are required during the construction of an **AVL tree**, we analyze the insertion of each element into the tree, maintaining the **balance factor** property $(-1, 0, 1)$ at every node. **Insertion sequence:** 35, 50, 40, 25, 30, 60, 78, 20, 28.

1. Insert 35

The tree has only one node, so no rotations are needed.

35

2. Insert 50

50 is greater than 35, so it becomes the right child of 35. The balance factor of 35 is $0-1=-1$, which is within the acceptable range $(-1, 0, 1)$. No rotations are required.

```

35
 \
 50
  
```

3. Insert 40

40 is greater than 35 but less than 50, so it becomes the left child of 50. The balance factor of 50 is $1-0=1$, and the balance factor of 35 is $0-2=-2$.

Imbalance at 35: This is a **Right-Left (RL)** case. Perform a **right rotation** on 50, followed by a **left rotation** on 35.

```

  40
 /  \
35   50
  
```

4. Insert 25

25 is less than 35, so it becomes the left child of 35. The balance factors of all nodes remain within the range. No rotations are required.

```

  40
 /  \
35   50
 /
25
  
```



/
25

5. Insert 30

30 is greater than 25 but less than 35, so it becomes the right child of 25. The balance factor of 35 becomes $2-0=2$, causing an imbalance.

Imbalance at 35: This is a **Left-Right (LR)** case.

Perform a **left rotation** on 25, followed by a **right rotation** on 35.

```

  40
 / \
30  50
 / \
25  35

```

6. Insert 60

60 is greater than 50, so it becomes the right child of 50. The balance factors of all nodes remain within the range. No rotations are required.

```

  40
 / \
30  50
 / \ \
25 35 60

```

7. Insert 78

78 is greater than 60, so it becomes the right child of 60. The balance factor of 50 becomes $0-2=-2$, causing an imbalance. **Imbalance at 50:** This is a **Right-Right (RR)** case. Perform a **left rotation** on 50.

```

  40
 / \
30  60
 / \ / \
25 35 50 78

```

8. Insert 20

20 is less than 25, so it becomes the left child of 25. The balance factors of all nodes remain within the range. No rotations are required.

```

  40
 / \
30  60
 / \ / \
25 35 50 78
 /
20

```

9. Insert 28

28 is greater than 25 but less than 30, so it becomes the right child of 25. The balance factors of all nodes remain within the range. No rotations are required.

```

  40

```

```

  / \
 30 60
 / \ / \
25 35 50 78
 / \
20 28

```

Hence, Total **Left Rotations** are 3

- 1 during RL at 35
- 1 during RR at 50
- 1 during LR at 25.

And Total **Right Rotations** are 2

- 1 during RL at 35
- 1 during LR at 25.

Q60 Text Solution:

Explanation:

Stack overflow is an internal interrupt.

The timer is an External interrupt.

The invalid opcode is a Machine check interrupt.

A superior call is a software interrupt.

Q61 Text Solution:

Ans : A

Using Armstrong's Axioms and the Union Rule to simplify the given set of functional dependencies (FDs) step by step:

1. Union Rule: Combine dependencies where possible, such as $EF \rightarrow G$ and $EF \rightarrow H$, into one:

$EF \rightarrow GH$

So, the FDs now become:

$F = \{A \rightarrow B, ABCD \rightarrow E, EF \rightarrow GH, ACDF \rightarrow EG\}$

2. Remove Extraneous Attributes:

- In $ABCD \rightarrow E$, B is extraneous. So, remove B from $ABCD \rightarrow E$, and we get: $F = \{A \rightarrow B, ACD \rightarrow E, EF \rightarrow GH, ACDF \rightarrow EG\}$

3. Further Simplification:

- E is extraneous in $ACDF \rightarrow EG$, So, remove E from $ACDF \rightarrow EG$, and we get: $F = \{A \rightarrow B, ACD \rightarrow E, EF \rightarrow GH, ACDF \rightarrow G\}$

4. Eliminate Redundancies:

- G is extraneous in $ACDF \rightarrow G$, because it's already implied by $EF \rightarrow GH$ and $ACD \rightarrow E$.
- we can see that $ACDF \rightarrow G$ can be derived using the remaining dependencies.

So, remove $ACDF \rightarrow G$, and the set becomes:

$F = \{A \rightarrow B, ACD \rightarrow E, EF \rightarrow GH\}$

5. Final Minimal Cover: The minimal cover of the functional dependencies is:

$F = \{A \rightarrow B, ACD \rightarrow E, EF \rightarrow G, EF \rightarrow H\}$

Q62 Text Solution:

Option- D(A statement that is executed automatically by the system as a side effect of



modification to the database)

Explanation: In a database, a trigger is a unique stored procedure as an automated process that activates in response to specific events. Like a notice, it alerts users to particular system-generated events such as:

INSERT: Addition of new record.

UPDATE: Modification to an existing record.

DELETE: Deletion of an added record.

Trigger operates without human help and also, maintains data integrity, enforces rules, and automates processes like logging or cascading changes whenever the database undergoes even a little modification.

Q63 Text Solution:

Option: **C(10)**

Explanation: Given page reference string,
4,3,2,1,4,3,5,4,3,2,1,5,

Using Least Recently Used (LRU) Page Algorithm with frame size 3, the number of page faults(miss) could be calculated as follows with the help of the below hit/miss table:

St ep	Refer ence	Fra mes	Miss/ Hit	Notes
1	4	[4]	Miss	Load 4 into memory.
2	3	[4, 3]	Miss	Load 3 into memory.
3	2	[4, 3, 2]	Miss	Load 2 into memory.
4	1	[1, 3, 2]	Miss	Replace 4 (LRU), and load 1 into memory.
5	4	[1, 4, 2]	Miss	Replace 3 (LRU), and load 4 into memory.
6	3	[1, 4, 3]	Miss	Replce 2(LRU) and load 3 into memory.
7	5	[5, 4, 3]	Miss	Replace 1 (LRU), and load 5 into memory.
8	4	[5, 4, 3]	Hit	4 are already in memory, with no replacement.
9	3	[5, 4, 3]	Hit	3 is already in memory, no replacement.
10	2	[2, 4, 3]	Miss	Replace 5 (LRU), and load 2 into memory.
11	1	[2, 1, 3]	Miss	Replace 4 (LRU), and load 1 into memory.
12	5	[2, 1, 5]	Miss	Replace 3 (LRU), and load 5 into memory.

Therefore, the total number of page faults= 10

Q64 Text Solution:

Options: **1.875 sec**

Explanation:

The tape has the following characteristics:

transport speed: 200 inches per second;

recording density: 1600 bytes per inch;

record size: 100 characters (usually one byte);

blocking factor: 10 records per block

The number of bytes per block= 1000 (

100 characters × 10 blocks per byte)

The number of blocks required is 600=

$$\frac{600000 \text{ bytes}}{1000 \text{ bytes per block}}$$

Length of tape required for 1 block=

$$\frac{\text{Number of bytes per block}}{\text{recording density}} = \frac{1000}{1600} = 0.625 \text{ inch}$$

Length of tape for 600 blocks=

$$\frac{\text{Length of tape for 1 block}}{\text{number of blocks}} = \frac{0.625}{600} = 375 \text{ inches}$$

Time required to write this length of tape=

$$\frac{\text{Length of tape}}{\text{transport speed}} = \frac{375}{200} = 1.875 \text{ seconds}$$

Note: As per UGC, the official answer is option A (2.0625 seconds).

However, the correct answer is 1.875 sec.

Q65 Text Solution:

Option A is Correct

The goal of PCP is to arrange the input in such a way that the string created by the denominator and the numerator is the same.

Firstly X,

List A	List B
1	111
10111	10
10	0

Start with 2nd row, $\frac{10111}{10}$

there is a remaining 111 in the numerator to satisfy this take 1st row,

$$\frac{10111}{10} \frac{1}{111}$$

now take again 1st row,

$$\frac{10111}{10} \frac{1}{111} \frac{1}{111}$$

by taking again the 3rd row,

$$\frac{10111}{10} \frac{1}{111} \frac{1}{111} \frac{10}{0}$$

X has a solution

Now Y,

List A	List B
10	101
011	11
101	011

Start from where the starting string matches, so start with 1st row,

$$\frac{10}{101} \text{ here remaining string in the denominator is 1.}$$

Now, take 3rd row,

$$\frac{10}{101} \frac{101}{011}$$

Only 2 Possibilities are there to make the match, either 1st Row or the 3rd row, Adding First Row it will become,

$$\frac{10}{101} \frac{101}{011} \frac{10}{101}$$



It gives an extra 1 in the denominator (101), which doesn't end up in same strings, and by taking the third row,

$$\frac{10}{101} \frac{101}{011} \frac{101}{011}$$

again in the last string in the denominator of last concatenation (011) will lead to further infinite concatenation, while concatenating further.

Hence, Y has no solution. So, In conclusion, Only PCP in X has a solution and PCP in Y has no Solution.

Q66 Text Solution:

Option- **D(All (A), (B) and (C) are not decidable)**

Explanation:

Property	Explanation	Decidability
(A) Finiteness	Since the Turing machine may operate endlessly, it is impossible to determine if a RE set is finite.	Undecidable
(B) Context Freedom	Since no method can determine if a RE language is context-free for all RE languages, it is impossible to say for sure.	Undecidable
(C) Emptiness	Since no method can determine if the Turing machine runs endlessly or if the set contains any items, it is impossible to determine whether a RE set is empty.	Undecidable

Q67 Text Solution:

Option-**A((A)-(III), (B)-(IV), (C)-(II), (D)-(I))**

Explanation:

List I	List II	Explanation
A. Activation Record	III. Subroutine Call	An activation record is created in response to a subroutine call in order to hold local state data during the subroutine's execution.
B. Location Counter	IV. Assembler	As it transforms assembly code into machine code, the assembler assigns addresses to data and instructions using the location counter
C. Reference Counter	II. Garbage Collector	To prevent memory leaks, the garbage collector manages memory using the reference counter, deallocating objects when their reference count drops to zero.
D. Address	I. Linking	To guarantee that the program's addresses are

List I	List II	Explanation
s Relocation	Loader	appropriately modified depending on where it is loaded, the linking loader is in charge of executing address relocation during the loading process.

Note: We have edited option A, as the options didn't match the answer.

The question was dropped by UGC

Q68 Text Solution:

Option- **D(All (A), (B) and (C))**

Explanation:

A) 4GT seeks to lower the effort needed for development. It does this by using higher-level languages, graphical tools, and frameworks to simplify complex tasks and speed up the process.

(B) 4GT also optimizes the use of resources, including developers' time and system resources (e.g., hardware, and software tools). It offers better tools and automation to manage and use resources efficiently during development.

(C) By increasing efficiency and decreasing the time and effort involved in development, 4GT lowers the cost by reducing the development time and increasing productivity.

Q69 Text Solution:

Option A is Correct

We are given the grammar:

$S \rightarrow SbS \mid a$

We are to consider the string "abababa" and determine which of the following statements are true:

(A) The string "abababa" has two parse trees.

(B) The string "abababa" has two leftmost derivations.

(C) The string "abababa" has two rightmost derivations.

The grammar allows us to generate strings consisting of 'a's and 'b's in a specific recursive pattern. The production $S \rightarrow SbS$ allows us to insert 'b's between 'S's recursively. Let's find different derivations of "abababa" to see if multiple parse trees and derivations exist.

For First Leftmost Derivation

- Start Symbol: S

- Apply $S \rightarrow SbS$

$S \rightarrow SbS$

- Replace the leftmost S with 'a':

$S \rightarrow abSbS$

- Replace the next S with 'a':



$S \rightarrow ababS$

- Replace the last S with 'a':

$S \rightarrow ababa$

- Final String: "ababa"

But this is too short. To get "abababa", we need more steps.

Corrected Leftmost Derivation for "abababa"

- Start Symbol: S

- Apply $S \rightarrow SbS$:

$S \rightarrow SbS$

- Apply $S \rightarrow SbSbS$:

$S \rightarrow SbSbS$

- Replace the leftmost S with 'a':

$S \rightarrow abSbS$

- Replace the next S with 'a':

$S \rightarrow ababS$

- Replace the last S with 'a':

$S \rightarrow abababS$

- Apply $S \rightarrow SbS$ again:

$S \rightarrow abababS$

- Replace the last S with 'a':

$S \rightarrow ababababS$

- Final String: "abababab"

Second Leftmost Derivation

- Start Symbol: S

- Apply $S \rightarrow a$:

$S \rightarrow a$

- Go back and apply $S \rightarrow SbS$ to the start symbol:

$S \rightarrow SbS$

- Replace the leftmost S with 'a':

$S \rightarrow abS$

- Apply $S \rightarrow SbS$ to the first S:

$S \rightarrow SbSbS$

- Replace the leftmost S with 'a':

$S \rightarrow ababS$

- Apply $S \rightarrow SbS$ to the first S:

$S \rightarrow abababS$

- Final String: "abababab"

Hence, There are at least two different leftmost derivations for "abababab". Statement (B) is true.

In the Similar Manner, We can construct two different rightmost derivations.

First Rightmost Derivation

- Start Symbol: S

- Apply $S \rightarrow SbS$:

$S \rightarrow SbS$

- Apply $S \rightarrow SbSbS$:

$S \rightarrow SbSbS$

- Replace the rightmost S with 'a':

$S \rightarrow SbSba$

- Replace the next S with 'a':

$S \rightarrow SbSbaba$

- Replace the last S with 'a':

$S \rightarrow SbSbaba$

- Apply $S \rightarrow SbS$ to the first S:

$S \rightarrow SbSbSbaba$

- Final String: "abababab"

Second Rightmost Derivation

- Start Symbol: S

- Apply $S \rightarrow SbS$:

$S \rightarrow SbS$

- Replace the rightmost S with 'a':

$S \rightarrow SbSa$

- Apply $S \rightarrow SbS$ to the first S:

$S \rightarrow SbSbSa$

- Replace the rightmost S with 'a':

$S \rightarrow SbSbSaba$

- Replace the last S with 'a':

$S \rightarrow SbSbSaba$

- Apply $S \rightarrow SbS$ to the first S:

$S \rightarrow SbSbSbSaba$

- Final String: "abababab"

Hence, There are at least two different rightmost derivations for "abababab". Statement (C) is true.

As Each unique derivation corresponds to a different parse tree. Since we have different derivations, there are at least two different parse trees for "abababab". Statement (A) is true.

Concluding the whole Description,
All Statements are True.

Q70 Text Solution:

Option- **A((B) and (C) only)**

Explanation: Alpha-beta pruning is used at all



levels of the tree and it usually gets more successful as depth increases. Many games use 6 or 8 levels of search depth. At these depths (6 and 8), there are enough branches to examine, and trimming significantly reduces the amount of computation memory.

Alpha-beta pruning is effective at all levels of the tree, but it gets more efficient at deeper levels (such as 6 and 8). Being the intermediate levels (6 and 8) therefore, option A is correct.

Q71 Text Solution:

Option- C((A) and (C) only)

Explanation:

A. $(\alpha \wedge (\beta \vee \gamma)) = (\alpha \wedge \beta) \vee (\alpha \wedge \gamma)$: The distributive property of logical conjunctions (AND) over disjunctions (OR) is demonstrated here. According to the distribution property, $A \wedge (B \vee C) = (A \wedge B) \vee (A \wedge C)$. In this instance, $(\alpha \wedge (\beta \vee \gamma))$ yields $((\alpha \wedge \beta) \vee (\alpha \wedge \gamma))$, which is true, when the distributive property is applied.

B. $(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha)$: This equivalency is contrapositive. The formula is $(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha)$. Since if β is false, then α must also be false, hence $\neg\beta \Rightarrow \neg\alpha$ must also be true. This equivalence is true if $\alpha \Rightarrow \beta$ is true.

Q72 Text Solution:

Option-C(Both (A) and(B) true)

Explanation:

$a * a * a = a * a * a$
 $a * (a * a) = (a * a) * a$ [since semigroup obeys assoc]
 $a * b = b * a$
 Now, since closure has to be satisfied, either $a * b = a$ or $a * b = b$
 1. If $a * b = a$,
 $b * b = (a * a)(a * a) = (a * a) * a = a * b * a = a * a = b$
 2. If $a * b = b$,
 $b * b = a * a * b = a * b = b$

Q73 Text Solution:

Option- D((A), (C) and (D) only)

Explanation: (A), (C) and (D) have a valid DFS order with the initial vertex 'A'.
 (B) does not have a valid DFS order because the edge from g to f is not visited.

Q74 Text Solution:

Option-A((A) and (B) only true)

Explanation:

A. Reflexive: For any element x, if $(x, x) \in (R)$, then a relation R is reflexive, meaning that x is connected to itself. In this instance: We determine if, for any x,

$$|x-x| < 0.0005$$

$$|x-x| = 0 \text{ which is less than } 0.0005$$

R is reflexive as the absolute difference between any number and itself is always 0 (less than 0.0005).

B. Symmetric: If for every x, y, if $(x, y) \in R$, then $(y, x) \in R$, a relation R is symmetric. In this instance, we must determine whether $|y-x| < 0.0005$.

Because the absolute value function is symmetric, the absolute difference $|x-y| = |y-x|$. Therefore, the relation is symmetric if $|x-y| < 0.0005$ and $|y-x| < 0.0005$

Q75 Text Solution:

Option-D(((A), B) and (C))

Explanation:

A. Memcached, a caching system, significantly enhances performance in Big Data ecosystems by managing high-throughput workloads and facilitating rapid data retrieval in systems handling large-scale, high-velocity data.

B. CouchDB stores data in JSON format and is intended for simple replication and scalability. It is crucial for Big Data applications because it can manage large datasets.

C. Graph databases like Infinite Graph are very helpful for big data applications that need to analyze linked data, like recommendation systems or social networks.

Q76 Text Solution:

Option: C((A)-(II), (B)-(III), (C)-(IV), (D)-(I))

Explanation:

List-I	List-II	Explanation
A. BCNF	II. It is not always dependency preserving.	The normalization process can sometimes result in the loss of dependencies that exist in the original relation.
B. 3NF	III. It removed transitive transparency.	Ensures that non-key attributes are not dependent on other non-key attributes.
C. 2NF	IV. It removes partial functional dependency.	The non-prime attribute is dependent on a part of the primary key (relevant to



		composite keys).
D. 4NF	I. It removes multivalued transparency.	Focus on eliminating multivalued dependencies (multivalued transparency), ensuring every determinant is a candidate key.

		operational details of a program component.	operational details of a program component and to restrict some features of a class or software component from being available to its customers.
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Q77 Text Solution:

Option: C((A)-(III), (B)-(I), (C)-(II), (D)-(IV))

Explanation:

	List-I		List-II	Explanation
A.	Localization	III.	Characteristic of software that indicates in a manner which information is concentrated in the program.	Describes the process of modifying software to accommodate various languages, geographical locations, or cultural norms.
B.	Packaging or binding of a collection of items.	I.	Encapsulation	Encapsulation is the process of enclosing code and data into a single unit and ensuring that sensitive information is concealed from users.
C.	Mechanism that enables a designer to focus on essential details of a program component.	II.	Abstraction	Allows the designer to concentrate on the most important aspects of a software component while paying little attention to the intricacies at a lower level.
D.	Information hiding.	IV.	Suppressing the	Capacity to obscure the

Q78 Text Solution:

Option: B((A)-(II), (B)-(III), (C)-(IV), (D)-(I))

Explanation:

	List-I		List-II	Explanation
A. Type 0		II.	Turing Machine	Most powerful, it can identify any language that can be computed, although it might not stop for every input (non-recursive).
B. Type 1		III.	Linear Bounded automata	Context-sensitive languages are recognized; their memory needs may be limited by the input duration.
C. Type 2		IV.	Pushdown automata	Identifies recursive structures (such as balanced parentheses) in context-free languages. To add more memory, they employ a stack.
D. Type 3		I.	Finite state automata	The most basic is the ability to recognize regular languages. There is no auxiliary memory (no tape or stack) in finite state automata.

Q79 Text Solution:

Option: D((A)-(IV), (B)-(II), (C)-(I), (D)-(III))

Explanation:

	List-I		List-II	Explanation
A. Ontological Engineering		IV.	Representing concepts, events, time, and physical concepts of different domains	Focuses on how knowledge about ideas, occurrences, and actual objects is represented and organized across a variety of fields.



B. Taxonomy Hierarchy	II. Organizing knowledge into categories and sub-categories	builds a hierarchy of concepts by classifying information and subgroups.
C. Inheritance	I. Organizing subclass relations	explains the connection between classes in which parent classes pass on traits or actions to their subclasses.
D. Probability Modes	III. Attach a number with each possibility	focusing on giving various options or outcomes numerical numbers, or probabilities.

Note: None of the options matches. Hence, we have edited the option D which is correct as per UGC.

Q80 Text Solution:

List I	Correct Match	Explanation
(A) Least frequently used	(II) Page replacement policy in cache memory	The Least Frequently Used (LFU) is a page replacement policy used in cache memory management.
(B) Critical Section	(III) Program section that once begun must complete execution before another processor accesses the same shared resource	The Critical Section is a program section that must complete execution before another process accesses the same shared resource, ensuring mutual exclusion .

(C) Loosely coupled multiprocessor system	(I) Memory is distributed among processors	In a Loosely Coupled Multiprocessor System , memory is generally distributed among processors .
(D) Distributed operating system organization	(IV) O/S routines are distributed among available processors	In a Distributed Operating System , O/S routines and tasks are distributed among available processors .

Correct Answer: C: (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

Q81 Text Solution:

List I	Correct Match	
(A) Firmware	(IV) ROM	Firmware is typically stored in ROM (Read-Only Memory) as it provides low-level control for hardware.
(B) Batch file	(II) ASCII format	A Batch file is usually in ASCII format , containing a sequence of commands to be executed by the command interpreter.
(C) Packing	(I) Number of logical records into physical blocks	Packing refers to the arrangement of logical records into physical blocks for efficient data storage and retrieval.
(D) Banker's Algorithm	(III) Resource allocation	Banker's Algorithm is used for



		resource allocation in operating systems to prevent deadlock.
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Correct Answer: C: (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

Q82 Text Solution:

List I	Correct Match	Explanation
(A) DES	(III) Key size - 56	DES (Data Encryption Standard) uses a key size of 56 bits.
(B) AES	(I) Key size - 256	AES (Advanced Encryption Standard) commonly uses key sizes of 128, 192, or 256 bits.
(C) 3 DES	(IV) Key size - 168	3DES (Triple DES) uses a key size of 168 bits, which is an enhancement over DES.
(D) RSA	(II) Key size - 1024	RSA typically uses key sizes of 1024, 2048, or higher for encryption.

Correct Answer: B: (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

Q83 Text Solution:

List I	List II	
(A) BIND	(II) Give a local address to a socket	The bind function assigns a local address to a socket.
(B) LISTEN	(III) Show willingness to accept connections	The listen function indicates the socket is ready to accept connections (shows willingness).
(C) ACCEPT	(I) Block the caller until a	The accept function blocks

	connection attempt arrives	the caller until a connection attempt arrives.
(D) SOCKET	(IV) Create a new point	The socket function creates a new communication endpoint.

Correct Answer: B: (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

Q84 Text Solution:

List I	Correct Match	Explanation
(A) Physical layer	(II) Make and break connections, define voltages and data rates, convert data bits into electrical signal	The Physical layer is responsible for making and breaking connections, defining voltages, and converting data bits into electrical signals.
(B) Data link layer	(III) Synchronization, error detection, and correction, assemble outgoing message into frames	The Data Link layer manages synchronization, error detection, and correction, and assembles outgoing messages into frames.
(C) Network layer	(I) Routing of the signals, divide the outgoing message into packets, act as network controller for routing data	The Network layer handles routing of signals, dividing outgoing messages into packets, and acting as a network controller for routing data.
(D) Presentation layer	(IV) It works as a translating layer	The Presentation layer functions as a translating layer, handling data format translation and encryption if necessary.



Correct Answer: D: (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

Q85 Text Solution:

List I	Correct Match	Explanation
(A) Breadth First Search	(II) $O(v+E)$	BFS has a time complexity of $O(v+E)$
(B) Rabin-Karp Algorithm	(III) $\theta((n-m-1)m)$	Rabin-Karp algorithm has a time complexity of $\theta((n-m-1)m)$
(C) Depth-First Search	(I) $\theta(v+E)$	DFS is represented by $\theta(v+E)$ in advanced analyses.
(D) Heap sort (worst case)	(V) $O(n \log n)$	Heap Sort has a worst-case time complexity of $O(n \log n)$
(E) Quick sort (worst case)	(IV) $O(n^2)$	Quick Sort has a worst-case time complexity of $O(n^2)$

Correct Answer is C: (A)-(II), (B)-(III), (C)-(I), (D)-(V), (E)-(IV)

Q86 Text Solution:

List I	Correct Match	Explanation
(A) Stack algorithm	(III) Page replacement	The Stack algorithm is used in Page Replacement algorithms, particularly in the Least Recently Used approach, in which the "stack" structure keeps track of the order of page usage.
(B) Elevator algorithm	(II) Disk scheduling	The Elevator algorithm is associated with Disk Scheduling . It operates similarly to an elevator, moving in one direction to serve requests until it reaches the end, then reversing direction
(C) Priority scheduling	(IV) CPU scheduling	The Priority Scheduling

algorithm		algorithm is used in CPU Scheduling , where processes are assigned priorities, and the CPU is allocated based on the priority level of each process.
(D) Havender's algorithm	(I) Deadlock	Havender's algorithm is related to Deadlock Prevention . It enforces conditions to prevent deadlock in concurrent systems by ensuring that processes follow a safe sequence.

Correct Answer: A: (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

Q87 Text Solution:

Statement I: Vertex-cover is a polynomial time 2-approximation algorithm.

True: The Vertex Cover problem has a 2-approximation algorithm that operates in polynomial time. In this algorithm, we find a vertex cover within twice the optimal size, it is called a 2-approximation algorithm.

Statement II: TSP-tour is a polynomial time 3-approximation algorithm for the Travelling Salesman Problem (TSP) with the triangle inequality.

False: For the Travelling Salesman Problem (TSP) with the triangle inequality, there exists a 2-approximation algorithm, not a 3-approximation. The triangle inequality allows for a 2-approximation approach (often using algorithms based on the Minimum Spanning Tree (MST) or Christofides' algorithm). Option A is the correct answer.

Q88 Text Solution:

Statement I: Conservative 2PL is a deadlock-free protocol.

True: In the Conservative Two-Phase Locking (2PL) protocol, all the locks a transaction needs are acquired before the transaction begins. If a transaction cannot acquire all the necessary



locks, it waits and does not acquire any locks, thus avoiding deadlock.

Statement II: Thomas's write rule enforces conflict serializability.

False: Thomas's write rule is used in timestamp ordering to ignore certain outdated write operations, thus increasing concurrency. Thomas Write Rule does not enforce Conflict Serializability but rejects fewer Write Operations by modifying the check Operations for $W_item(X)$.

Statement III: Timestamp ordering protocol ensures serializability based on the order of transaction timestamps.

True: The timestamp ordering protocol assigns a unique timestamp to each transaction and ensures serializability by executing transactions in the order of their timestamps.

The correct option is B.

Q89 Text Solution:

Statement I: Composite attributes cannot be divided into smaller subparts.

False: Composite attributes are, by definition, attributes that can be divided into smaller subparts. For example, a "Name" attribute can be divided into "First Name" and "Last Name."

Statement II: Complex attribute is formed by nesting composite attributes and multivalued attributes in an arbitrary way.

True: Complex attributes are formed by combining composite attributes and multivalued attributes in various ways.

Statement III: A derived attribute is an attribute whose values are computed from other attributes.

True: A derived attribute is computed based on the values of other attributes. For example, "Age" might be derived from a "Date of Birth" attribute. Option D is correct.

Q90 Text Solution:

Statement I: Working from the general to the specific.

True: In the top-down approach, development starts with a high-level design, focusing on the general structure before moving into specific details. This is a characteristic of the top-down approach.

Statement II: Postpone the minor decisions.

True: The top-down approach involves focusing on the overall structure first while deferring minor or detailed decisions until later in the development process. This aligns with the principle of working from general concepts to specific implementations.

Statement III: A systematic approach.

True: The top-down approach is systematic, as it follows a structured process of breaking down a complex problem into manageable parts. This approach ensures that each part is addressed in a logical order.

Option C is correct.

Q91 Text Solution:

Statement I: LALR parser is more powerful than canonical LR parser

- **False:** Canonical LR parsers (also known as LR(1) parsers) are more powerful than LALR parsers because they can handle a larger class of grammars. Canonical LR parsers have separate states for each lookahead, which makes them more powerful than LALR parsers, which combine states with the same core but different lookaheads to reduce the number of states.

Statement II: SLR parser is more powerful than LALR

- **False:** LALR parsers are more powerful than SLR parsers. LALR parsers handle a larger set of grammars than SLR parsers because they use lookahead symbols more effectively. SLR parsers use a simple lookahead mechanism, which makes them less powerful than LALR parsers.

Option C is correct.

Q92 Text Solution:

Statement I: CFL is closed under homomorphism

- **True:** A homomorphism is a function that maps symbols from one alphabet to strings over another alphabet. Context-free languages are closed under homomorphism. This means that if L is a CFL, applying a homomorphism to L will result in another CFL.

Statement II: CFL is closed under complement

- **False:** Context-free languages are not closed under complement. This means that if L is a CFL, the complement of L may not necessarily be a CFL. This is a property of CFLs.

Option A is correct.

Q93 Text Solution:

Statement X

$$X: a \vee (b \wedge (a \vee c)) = (a \vee b) \wedge (a \vee c)$$

This is the **Distributive Law** in Boolean algebra, which distribute $a \vee$ over $b \wedge (a \vee c)$:

$$a \vee (b \wedge (a \vee c)) = (a \vee b) \wedge (a \vee c)$$

For X to be true, this distributive property should hold.

Statement Y

$$Y: a \wedge (b \vee (a \wedge c)) = (a \wedge b) \vee (a \wedge c)$$



This is also a form of the **Distributive Law** in Boolean algebra, which distributes $a \wedge$ over $b \vee (a \wedge c)$:

$$a \wedge (b \vee (a \wedge c)) = (a \wedge b) \vee (a \wedge c)$$

For Y to be true, this distributive property must hold as well.

Analysis:

The expression $a \vee (b \wedge c) = (a \vee b) \wedge c$ does not generally hold in Boolean algebra for arbitrary values of a, b, and c. This expression would only be true under specific conditions, such as if c implies a, or if a and b have particular truth values.

Options

1. Option A: X is true

- X is true, then the expression $a \vee (b \wedge c) = (a \vee b) \wedge c$ is satisfied. However, the expression does not depend solely on the truth of X, as it requires specific conditions on a, b, and c.

2. Option B: Y is true

- Y is true, then the expression $a \vee (b \wedge c) = (a \vee b) \wedge c$ is satisfied. The truth of Y alone is not sufficient to satisfy the expression.

3. Option C: Both X and Y are true

- Both X and Y are true, then the expression $a \vee (b \wedge c) = (a \vee b) \wedge c$ is satisfied. This expression does not depend on the truth of X or Y but on the specific values of a, b, and c.

Q94 Text Solution:

A good Software Requirements Specification (SRS) should have characteristics like **completeness, consistency, and clarity** to ensure that the requirements are well-defined, unambiguous, and understandable. These characteristics help in accurately defining what the software should do.

Reliability, however, is not a characteristic of a good SRS document itself. Reliability is a characteristic of the **software product** being developed, not of the requirements specification document.

Option D is correct answer.

Q95 Text Solution:

Assertion (A): $\bar{p}$

This assertion states that p is false, or logically, it is the negation of p.

Reason (R): $(r \rightarrow \bar{q}, r \vee s, s \rightarrow \bar{q}, p \rightarrow q)$

$r \rightarrow \bar{q}$: If r is true, then q is false.

$r \vee s$: Either r or s is true.

$s \rightarrow \bar{q}$: If s is true, then q is false.

$p \rightarrow q$: If p is true, then q is true.

These implications do not directly relate to the truth of p. The Reason (R) involves conditions about r, s, and q, but it does not provide a why p should be false.

- **Assertion (A):** $\bar{p}$ is a statement that p is false, which can be considered independently true or false based on the truth value of p.
- **Reason (R):** The statements in (R) do not provide a direct explanation for why p should be false. Therefore, (R) does not serve as a correct explanation for (A).

The correct option is: B.

Q96 Text Solution:

The correct answer is **option D**.

A logical error is a bug in a program that causes it to produce incorrect or unexpected results.

"Using commas instead of two required semicolons in a for loop header" is not a logical error as it is a syntax error and the rest of the three are logical errors.

Q97 Text Solution:

The correct answer is **option A**.

Load-and-go assemblers generate their object code in memory for intermediate execution. No object program is written out, no loader is needed.

Assertion (A): A load-and-go assembler avoids the overhead of writing the object program out and reading it back in.

This statement is true. A load-and-go assembler directly loads the program into memory and executes it without creating an intermediate object file, thus eliminating the overhead associated with writing to and reading from storage.

Reason (R): This can be done with either a one-pass or two-pass assembler.

This statement is also true. Both one-pass and two-pass assemblers can be designed as load-and-go. One-pass assemblers also avoid the overhead of an additional pass over the source program.

Q98 Text Solution:

The correct answer is **option C**.

True: A) All function calls in C pass arguments using call by value. In C, function arguments are always passed by value, meaning that a copy of the argument is made.

True: B) Call by reference enables a called function to modify a variable in the calling function. Call by reference allows the called function to access and modify the original variable in the calling function.



False: C) Call by value is always more efficient than call by reference. While call by value can be more efficient in some situations (especially with small data types), it is not always more efficient than call by reference. For large data structures, passing by reference can be more efficient because it avoids copying large amounts of data.

True : D) Programmers use pointers and indirection operations to simulate call by reference. In C, pointers are used to pass the address of a variable, allowing the called function to modify the variable's value, effectively simulating call by reference.

Q99 Text Solution:

The correct answer is **option A**.

Correct: Statement I "Grandparent is a parent of one's parent."

This statement is correct because a grandparent is defined as the parent of a person's parent.

Correct: Statement II "First Order Predicate Logic (FOPL) representation of the above statement is

$\forall g, c \text{ grandparent}(g, c) \Leftrightarrow \exists p \text{ parent}(g, p) \wedge \text{parent}(p, c).$ "

To analyze this,

The left side states that for all individuals g and c, g is a grandparent of c.

The right side states that there exists an individual p such that g is a parent of p and p is a parent of c.

This representation correctly captures the relationship defined in Statement I. Therefore, Statement II is also correct.

Q100 Text Solution:

The correct answer is **option A**.

(Correct) Statement I: It describes a 20-bit microinstruction code format with specified bit allocations for different fields (F1, F2, F3, CD, BR, AD). The bits are given as 3 bits for each of F1, F2, and F3, 2 bits for CD, 2 bits for BR, and 7 bits for AD. This format appears to be suitable for a microinstruction code, and the allocation seems reasonable for controlling various operations and addressing. It accurately describes a microinstruction code format.

(Correct) Statement II: It claims that the instruction represented in the above format can perform a branch in 4 conditions. Given that the CD field is 2 bits, it can represent ($2^2 = 4$) different conditions (00, 01, 10, 11). This aligns with the statement about branching in 4 conditions.

